

Start Driving

The ground rules of traffic

- The first paragraph of the law that regulates the rules for all road users is about attitudes. When you read the paragraph, it is easy to feel that you have only obligations and no rights. This interpretation is correct. With the fact that we do not have any rights. The law compels us road users to be careful.

The law summarizing words:

- Care and caution.
- Consideration.
- Do not hinder and interfere.

The law applies to all roads and everywhere. All that is not road is in a legal sense terrain and there the law also applies. Most traffic rules thus apply everywhere.

You as a road user shall show all the care and attention required so that an accident does not happen. You should show enough consideration that nobody gets hurt by your actions. You must not hinder or interfere with anyone.

Provisions for all road users, Basic Provisions, § 1 (traffic regulations - 1998: 1276):

- To avoid accidents road users shall observe the care and caution required in the specific situation. Road users should show consideration towards children, the elderly, school patrols and persons perceived to have a disability or illness that impedes them in traffic.
- A road user shall act so that he or she does not unnecessarily hinder or interfere with other traffic.
- Road users should be considerate to those who live or reside along the road.
- An off road user must adapt their route and speed, as well as their means of transport so that humans and animals are not unnecessarily disturbed and so that damage to someone else's land or vegetation is avoided.

Defensive driving

The correct answers to many problems in traffic are often "Drive defensively". But it is important that this response does not become a cliché for you. You really need to drive defensively.

What is defensive driving? Here are some examples:

- Have plenty of time.
- Drive with margins.
- Drive clearly so others can see your intentions.
- Expect others to do wrong.
- Drive smoothly.
- Drive in an understandable way and show others that you intend to give way when you must give way or have obligation to stop.
- Keep good attention, both forward and backward.
- Be prepared for the unexpected!

You as the driver have the responsibility to do everything possible to avoid an accident. You can never blame someone else, you are always responsible. Driving a vehicle is not a right for which others should conform to you. You always have a great responsibility, even if others do "wrong" in traffic.

Planned driving

Planning your driving, your journey, starts already at home, even before you start driving. Give yourself plenty of time to avoid being stressed.

During the actual drive, look far ahead and be prepared for the unexpected. Think about your choices. Dedicate yourself to "driving". Be focused, avoid using your mobile phone or other equipment that may distract you.

When you plan your driving well, you have plenty of time and become less stressed. You therefore drive safer and probably also more environmentally friendly.

When planning fails, keep in mind that it is better to arrive a little late than not at all.

Obstacle

Parking on a sidewalk is never allowed. The sidewalk is for pedestrians and do not forget that there are pedestrians who need to get around with aids such as a walker or wheelchair. The fact that there is then a car on the sidewalk makes it very difficult.

If someone parks so that you can not drive past in a normal way, you may drive past the car, if required, even on the wrong side.

Different types of roads

A road consists of several parts, and to understand how different traffic rules work, you should know the different concepts.

Road (Väg)

This is the whole road.

Roadway (Körbana)

This is the part of the road that is typically used by vehicles. Cycle lanes, footpaths, horse paths, and hard shoulders are not included.

Lane (Körfält)

This part of the road is used for travel in one direction by four-wheeled vehicles. The lane is usually marked with road markings, but it does not have to be. If, for example, it is missing the center line and two four-wheelers can meet, this counts as two lanes.

Hard shoulder (Vägren)

This is part of the road outside the edges of the roadway.

Hard Shoulder

To the right of the roadway is usually a border line. The area outside this line is called the hard shoulder.

This is not a lane. It is an additional area that are primarily to increase safety on the road. It is often used by cyclists, moped riders and other slow-moving vehicles.

Since pedestrians use the hard shoulder, you should absolutely not drive on it in the dark. You may use the hard shoulder to facilitate other traffic to overtake you. You have a huge responsibility when you do this. If an obstacle appears on the hard shoulder, you must give way to traffic on the roadway according to the exit rule.

You may also use the hard shoulder when you leave a bigger country road. This can improve security but remember that if you are not absolutely sure that it is free, do not use the hard shoulder.

Main road

On the hard shoulder on a main road you are allowed to stop but not park.

Lane

A lane is a longitudinal portion of the road that is marked with lines. Even if lines are missing, it may still count as several lanes if several four-wheel vehicles can fit next to each other.

When there are lane markings, you must follow them. It is important that you have good planning since late lane changes often can cause difficult situations in traffic. If you end up in the wrong place it is often safer to follow the traffic and turn in any suitable place.

Streets with three or more lanes

On a two-way street (a street with traffic in both directions) with three lanes, you may not use the lane on the far left. If the street has more than three lanes, you may not drive in the lanes to the left of the lane in the middle of the direction of travel. The reason is that those lanes are for oncoming traffic.

If the street is one-way, you can use all lanes.

Solid lines normally always divide the lanes. The above rules clarify cases where refuges or solid lines are missing. Snow can also hide the lines so the rules are required.

Roadway

The part of the road that normally is used for the circulation of vehicles.

Cycle lanes, pedestrian lanes, horse paths and hard shoulders are not included.

Start Practice Driving

Start and stop

One of the very first exercises in driving license training is to start and stop. This means that you should learn the basics of how to start correctly from a roadside and start driving. The emphasis is on routines regarding how you look around you, use the rear-view mirrors and give signs.

For the exercise to be effective you usually only drive away and after a short distance stop again. Then the emphasis is on a stop with good attention to the surroundings and without unnecessary jerks.

Be careful, there is a great risk that you will judge the speed of other traffic incorrectly and start with too small margins for other traffic and thus in the worst case may be hit.

Change driver

At the beginning of your education, you and your teacher often change places when the teacher may, for example, want to demonstrate. Carefully check other traffic so you do not forget about it and cause an accident.

Crawl driving

Manual shifting

Crawl driving means driving so slow in first gear that you on flat ground regulate the speed with the clutch. This means that if you depress the clutch pedal the car stops completely and if you raise the clutch pedal the car starts moving again. If it's an uphill you may also need to give some ground gas.

Automatic shifting

If you are driving an automatic transmission, you control the speed primarily with the brake pedal.

Attention

Most beginners lock their gaze too close to their own vehicle. This means that they do not get the field of vision they so desperately need to be able to drive safely.

Here are some guidelines to gain good attention :

- Look far ahead.
- Keep your gaze moving.
- Use your cars mirrors.

Distance assessment

As drivers, we are very dependent on our distance judgment working well. Unfortunately, we humans have rather limited abilities when it comes to judging the distance to, and speed of, other vehicles.

We need clues

In order to judge distances in a good way, we need clues in the form of things that we know the size of, for example, houses and trees. We have more difficulty judging the distance to an oncoming vehicle coming from the front than if it is coming from a connecting road.

It is more difficult in the dark and with poor visibility

We find it very difficult to judge distance and speed in the dark and bad weather.

Common mistakes

- You assume that the point where your vehicles meet is midway between where you are now and where the oncoming vehicle is
- You feel that the oncoming person is making a mistake and maintain your position when you meet someone on your own side of the road. Instead, move to the right, slow down, and use the road divider if there is one and you have a clear view.

Use of indicators

For the traffic to work well, you must inform others of what you intend to do by driving clearly and giving signs.

On the following occasions, you must use blinkers (give signs).

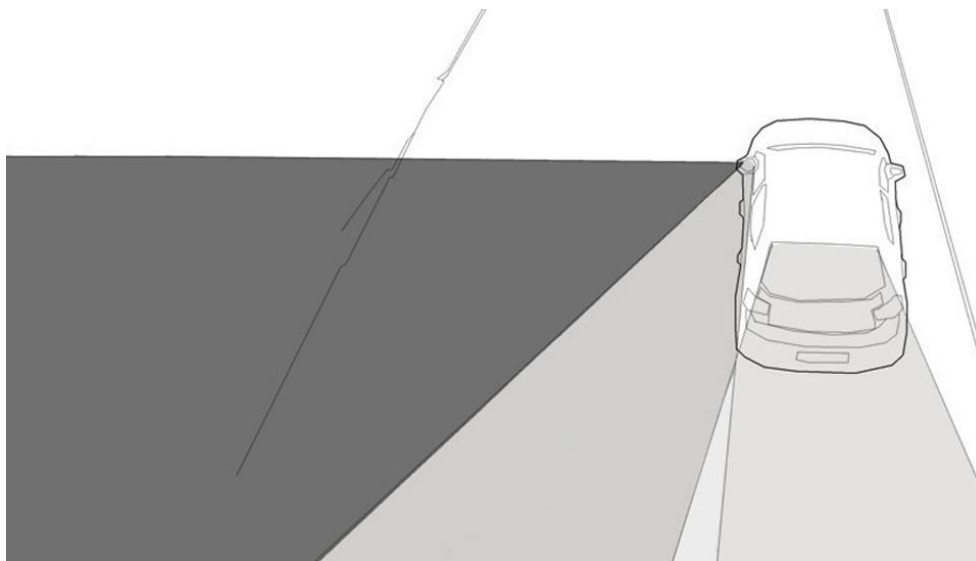
When you

- starting from the roadside
- turns on the road
- turning at an intersection
- changing lanes
- moves the car sideways so much that it can disrupt other traffic.

Note that you must signal when turning at an intersection. You must therefore signal even when driving in a lane with a lane arrow. You must signal to guide other road users, including pedestrians, and your signal must not be misunderstood.

Regardless of whether you give signs, you must always ensure no danger or accident occurs.

Blind spot



The blind spot is the area you have at your back on the left and right sides that is not visible in the car's rearview mirrors.

Look at the picture. The dark gray area shows where the blind spot is on your left when you start from a roadside. You have to turn your head to check the blind spot. Missing a vehicle in the blind spot can lead to a serious accident.

Sit in a car and look only in the rearview mirrors. Ask someone to stand behind the car and then go along the sides past the car at about the same distance as a car that drives by you. You will discover that you cannot see the person all the time. This simply shows where you have blind spots.

When you are in the blind spot

When you pass another vehicle you should be particularly alert in the area where you know you are in the other driver's blind spot.

Think about the environment

Think about the environment as you practice driving. When planning an exercise and reading what to do, you should not have the car idle.

Gear

Clutch

Manual shift only

A car with manual shifting has a clutch pedal which is often called a clutch.

When the clutch pedal is fully released and gear is engaged, the power from the motor to the wheels driving the car is transmitted through something called the slat plate. This causes the car to move.

When you push down on the clutch pedal all the way down, the power from the engine is released so that the wheels can no longer drive the car.

When the clutch pedal is between these two end positions, the car can either stand still or move slowly forward depending on how far down you pedal down the clutch pedal. This mode is usually referred to as "pull mode".

Disengaging

Depressing the clutch pedal to the bottom is usually called disengaging the car.

Automatic transmission

If your car is equipped with an automatic transmission, you do not shift gears yourself. It is enough to accelerate and brake. Normally, there is not even a clutch pedal since the "change" is handled automatically.

It is ok to practice driving with a car that has automatic transmission. It is even so that it can be an advantage in the beginning of training since you have more time to engage in traffic, learn to look far ahead, using mirrors and more. If you practice driving with an automatic transmission, you should use the right foot for both the brake and gas. Never use the left foot!

Function

An automatic transmission operates by means of gears of different sizes and hydraulic oil. The oil is usually only controlled at a workshop in connection with servicing the car.

Check in your car's owner's manual what the different positions on the gear lever mean. Most cars usually have the modes:

- D (drive) to drive.
- R (reverse) to reverse.
- N (neutral) to roll the car, e.g. in a car wash.
- P (park) to park the car.

Driving license with conditions

You can do your driving test with an automatic car. You will receive a license with condition "Automat", which entitles you to drive only cars equipped with automatic transmission.

Positioning

When you start learning to drive, you should of course drive on the right side. Sometimes it is difficult in connection with turning when you as a beginner have a lot to think about because you have to blink, brake, shift and, watch out for. If you shift manually, you have a lot to do with choosing the right gear and more.

To avoid getting on the wrong side of the middle of the road after the turn, keep in mind the following:

- Be careful not to turn too late when you turn right. This makes it easier for you not to get a turn that comes in on the wrong side of the new street.
- Be careful not to turn too early when turning left. One of the most common problems associated with a left turn is that you "cut across" in the turn.

The right gear when turning

When training on downshifting, it is important that you always have the right speed before shifting. There are generally two methods of getting the right speed:

- Use the engine brake. You do this by releasing the gas in good time. Then the speed drops by itself. This way of driving is best for the environment as you avoid a lot of braking and the driving becomes soft and comfortable while saving fuel.
- Use the foot brake. You do this if you need to slow down faster.

Manual shift only

If you need to slow down really fast, you may also need to depress the clutch while braking. Normally you should do this only when inserting the new gear.

Engine brake

Engine brake with internal combustion engine

"Engine braking" means the braking effect that occurs when the throttle is released. This effect is less in a car with automatic transmission than in cars with manual transmission. When driving with automatic transmission, you can usually release the gas earlier than you would with a manual gearbox. This is because the braking effect is smaller.

Engine brake as much as possible. Plan the drive and slow down by letting off the gas in time - and let it off completely to reduce the fuel supply and thus save fuel. Shift down just before the engine speed is reduced to 1200-1300 rpm. If you do not have a tachometer on the car, you will learn to listen to the engine sound.

A major advantage of engine braking is that the fuel consumption of a modern car goes down to almost zero when the fuel supply is cut off while you are slowing down.

Note! When you "engine brake" it does not turn on the vehicle's brake lights. Therefore, if you have someone behind you, especially if someone is close behind you, tap on the brake pedal and, the brake light turns on the car. Thus, you make it easier for the people that the attention that you lower the speed.

"Engine brake" with electric car

If you have an electric car, the "engine brake" works differently. Much depends on how you have "set up the car". On most electric cars, you can choose whether you want a lot or a little "engine braking" when you let off the gas.

"Engine brake" on an electric car actually refers to regeneration of the energy. Regeneration is to recover kinetic energy. When you release the gas pedal, the kinetic energy is converted into electrical energy and stored in the car's battery, while the speed drops noticeably. There are electric cars where you can drive completely without using the brake pedal.

How the brake lights work during regeneration is different between different cars. You must check how your car works in the car's owner's manual. Keep in mind that if you are not

completely sure how your car works, it is good to use the brake pedal when there is someone very close behind you when you intend to brake and make a turn.

Driving Theory

Braking

Shifting, breaking

15 / 91

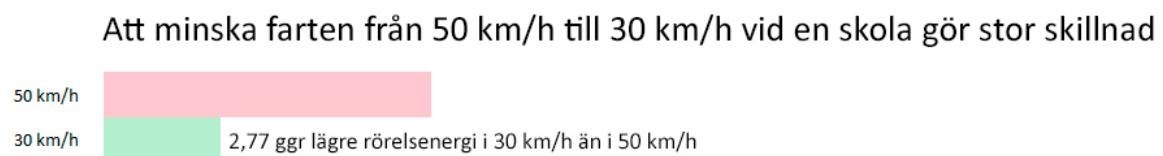
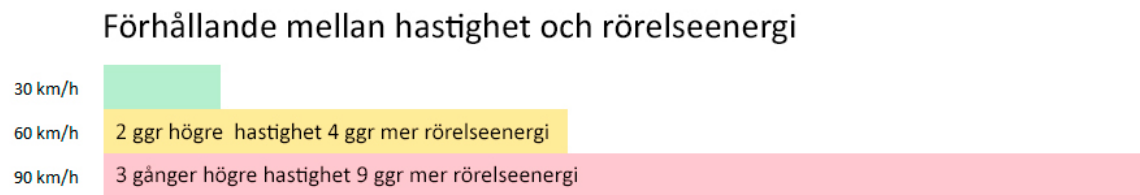
The basics of speed

When driving, you have to keep a certain speed. Thus, "speed" cannot be avoided. Increasing the speed increases the risks more than you might think. It is therefore important that you become proficient at braking the car.

Great powers - use seat belt

Already at a speed of 10 km/h you can not stay in the seat without a seat belt during a hard braking.

Kinetic energy



Click on the picture to enlarge378

The word **kinetic energy** is quite descriptive, it is the energy of an object that is in motion.

Kinetic energy is affected by weight and speed. Kinetic energy increases fast with higher speed. The increase is so great that when you double the speed you actually have quadrupled the kinetic energy and thus the braking distance to the same extent. If you triple the speed, you get nine times as much kinetic energy

The kinetic energy increases according to the following:

- $2 \times 2 = 4$
- $3 \times 3 = 9$
- $4 \times 4 = 16$
- ...

The reverse also applies, if you halve the speed, you have reduced the kinetic energy by 4 times. This is why small reductions in the speed at, for example, a school, result in a large reduction in the force of the crash in the event of an accident.

It is very important that you as a driver understand the importance of this law of nature.

It is the braking distance that is affected by the kinetic energy. And since the braking distance is part of the **stopping distance**, this is also affected.

Example when increasing speed

When you double the speed the length of the braking distance quadruples.

- If you at 30 km/h have 6 meters braking distance, at 60 km/h you have 24 meters ($6 \times 4 = 24$)

When you triple your speed the length of the braking distance is 9 times longer.

- If you at 30 km/h have 6 meters braking distance, at 90 km/h you have 54 meters ($6 \times 9 = 54$)

Example when reducing speed

The relationship between speed and braking distance is the same when you reduce speed.

- If you reduce your speed by half it reduces the length of the braking distance four times.
- If you reduce speed to one-third it reduces the length of the braking distance by nine times.

It is easy to misjudge

You often have a longer braking distance than you think. Do not ignore the importance of small reductions in speed, they mean a lot to the braking distance of the journey.

Stopping distance

Stopping distance refers to the distance the vehicle travels from the time you find that you need to stop until the vehicle has stopped.

The stopping distance consists of two parts, the **reaction distance + braking distance**.

Reaction distance

The distance the car goes from when you discovered the obstacle until you start to slow down. The reaction distance is affected by your speed and reaction time (ability to react).

Reaction time

Already at 50 km/h you move about 15 meters per second. This corresponds to about three car lengths. In normal cases, the reaction time is one second. If something unexpected happens when driving at 50 km/h, you will move three car lengths before you even start to slow down.

Braking distance

The distance the car goes from when you started braking until the vehicle is stationary. The braking distance is affected by many factors, including speed and road conditions. The braking distance can be up to ten times longer when there are bad conditions.

Tips for converting km / h to m / s

Loading image...

$$\cancel{30} \text{ km/h} = 3 * 3 = 9 \text{ m/s}$$

Click on the picture to enlarge3116

Sometimes it can be good to know how many meters per second you move when driving at a certain speed.

The terms used are

- km/h means kilometers per hour, **h** stands for " hour". The teaching material always use km/h to account for speed.
- m/s means meters per second

Simple method

Below is a method that gives a result that is slightly too much but the results are sufficiently accurate to be fully useful.

Delete the last zero and multiply by three , some examples:

- 30 km/h is $3 * 3 = 9 \text{ m/s}$
- 50 km/h is $5 * 3 = 15 \text{ m/s}$
- 70 km/h is $7 * 3 = 21 \text{ m/s}$
- 100 km/h is $10 * 3 = 30 \text{ m/s}$

Exact method

The exact measurements for comparison :

- 30 km/h, equal to 8.33 m/s , the exact formula is $30 / 3.6$
- 50 km/h, equal to 13.89 m/s , the exact formula is $50 / 3.6$
- 70 km/h, equal to 19.44 m/s , the exact formula is $70 / 3.6$
- 100 km/h, equal to 27.78 m/s , the exact formula is $100 / 3.6$

For the interested

- 1 km is 1 000 meters
- 1 hour is 3 600 seconds
- 1 000 meters / 3 600 seconds is 0.27 meters per second
- If you divide both numbers by 1,000, you get $1 / 3.6$

Study Questions

Study questions are answered using the information above including the **Read More** words available. External links are only in-depth.

Loading image...



Click on the picture to enlarge141

About how many meters do you move in one second?

If I drive 30 km/h I move about 9 meters

If I drive 50 km/h I move about 20 meters

If I drive 70 km/h I move about 28 meters

If I drive 90 km/h I move about 32 meters

Loading image...



Click on the picture to enlarge302

If you double your speed, how much your kinetic energy increase?

It is doubled

It is trippled

It is quadrupled



Click on the picture to enlarge305

What affects kinetic energy?

My cars weight and vehicle speed

My cars speed and the road conditions on the road I drive on

My cars weight is the only thing that affects kinetic energy

My cars speed is the only thing that affects kinetic energy



Click on the picture to enlarge314

This is about reaction time, what is correct?

There is no difference between young and old drivers reaction time

Young drivers have more than two seconds faster reaction time than older

In complicated situations the experienced driver have shorter reaction time than the inexperienced



Click on the picture to enlarge124

This is about kinetic energy, which statement is correct?

The kinetic energy has no meaning in accidents

The kinetic energy mostly affects when I am driving in curves

The kinetic energy increases four times when you double your speed

Downhill and uphill

Driving on a slope

16 / 91

Drive downhill and uphill

This step is about being able to drive on a slope as well as how you can use the parking brake as an aid.

The technique for starting on an uphill or downhill slope differs and this mainly depends on whether you drive a car with a manual gearbox or automatic gearbox.

Start uphill

Manual shifting

To start in a good way uphill you first need to get the car ready to go. You must be stationary on the slopes with the help of the gas and the car's clutch engagement.

- **Slight uphill** - On a slight uphill slope, you can often find the car's clutch engagement point without using the parking brake. Once you have found the pull mode, make a normal start by checking if it is free of traffic, giving signs and driving away.
- **Steep uphill** - If there is a steep uphill slope you can search the clutch engagement point before releasing the parking brake. Then, when you are stationary on the slope, you make an ordinary start.

Automatic shifting

With automatic shifting, you usually keep the car still with the foot brake. When you release the foot brake to accelerate, the car may roll backward or slowly forwards depending on the load, incline and, car you have. Some cars with automatic transmission, for example, have something called **Auto Hold** or **Hill Hold**, which means that the car holds the brake for a certain time or until you press the throttle. If you are unsure of how the car you are driving works, look in the instruction manual or test yourself in a safe place. When you are ready to go, see if it is free, give the signal and drive away.

Start downhill

Manual shift

When you start downhill you sometimes can use the slope to get the car in motion so that you then can start in second gear.

- **Start in second gear** - You make the car ready to drive by switching into second gear and keep the car stationary with the brake. Then you have a look around in the regular way, if it is free you turn on your turn signal and let the car start by releasing the brake. This way you will not have to use first gear.

Automatic shifting

With automatic shifting, you usually keep the car still with the foot brake. When you are ready to go, see if it is free, give the signal and drive away by releasing the brake and giving gentle throttle.

Study Questions

Study questions are answered using the information above including the **Read More** words available. External links are only in-depth.



Click on the picture to enlarge1037

You are on an uphill climb. What best describes the interception when driving a car with a manual shift?

I brake just before the car starts to roll backward

I take an opportunity and drive away when it is free

I get the car to a standstill on an uphill slope with the help of gas and clutch



Click on the picture to enlarge5518

What is true of the parking brake?

The parking brake always work on all four wheels and is a good alternativ for the standard brake

The parking brake operates mostly on two wheels and has a limited ability to stop a moving car

The parking brake is always a lever between the cars front seats



Click on the picture to enlarge1040

Your car has a manual gearbox. You have been parked on a slight uphill slope. You're going to drive away. Which option is correct?

I crawl drive forward and when it is free I give signs and drive away

I turn on my turn signal, search for the car's clutch engagement point, look around and drive away

I hold the car stationary at the cars clutch engagement point, look around, and when it is free, I use my turn signal and drive away



Click on the picture to enlarge1041

How do you best start on a steep downhill when your car has a manual gearbox?

I can often start in second gear as the car roll to a start by itself

I do not need to use my turn signal since I let the car roll to a start

I always start in first gear

Maneuvering the car

This step is about reversing, parking and turning.

When you are practicing driving, it is important to operate the car.

Turning radius



Click on the picture to enlarge1182

"Turning radius" usually, in regards to drivers education, mean the fact that the cars front end "takes up space" when turning.

Imagine that you are standing between two cars in a parking lot and should reverse out. If you then turn right away with the steering wheel, the cars front end, in all probability, will hit the car next to you. This is something you always have to think about, especially when reversing.

Heavy vehicles

A turning radius can also mean that heavy vehicles need room to manage their turn. See the picture where the bus is dependent on that no one stop or park where there is a yellow zigzag markings.

Reverse



Click on the picture to enlarge565

When you are going to go in reverse, it is often difficult to see out the back. This makes going in reverse a fairly dangerous maneuver. Since you usually go in reverse towards the direction of travel, you need to have very good supervision.

The law is clear:

- You may not reverse if it can endanger or hinder others.

Attention

Always check carefully that no one is behind the vehicle when you start to back up. This may sound obvious but it happens that we drive into different objects and in the worst cases even backs into people. Remember that a child hiding behind the car is very difficult to detect. You may use the mirrors when reversing to control your direction but essentially you should look out the cars rear window. Remember that you must turn now and then to see the vehicle both sides. When you reverse, stop when there are other vehicles. This reduces the risk of misunderstanding. If you take help of someone from the outside, you can not unload the responsibility on this person. It is always you who are responsible for anything that happens when you reverse.

Aid

If your car has a rear-view camera, you should use this. Look at it before you start to reverse. The camera helps you see obstacles immediately behind the car. You should stand still when you look into the camera, it is only an aid.

One-way street

If you intend to turn by backing up at an intersection, you may not go in reverse into a one way street.

Study Questions

Study questions are answered using the information above including the **Read More** words available. External links are only in-depth.



Click on the picture to enlarge120

You and your teacher get help from a person outside the car when you practice reversing. You collide with another car, what is correct?

The one who helps us never has any responsibility

The one who helps us shares the responsibility with my instructor

The one who helps us has the full responsibility



Click on the picture to enlarge242

Which statement is correct when reversing?

It is easy to detect obstacles behind the car when reversing

It is easy to miss children behind the vehicle when reversing

It is easy to see all approaching from the sides when reversing



Click on the picture to enlarge339

It is dangerous to reverse, which of the following risks is greatest when there are small children nearby?

Children usually realize that I'm reversing but I might look at them too much and miss something else

The children are very fickle and can indulge in playing "hide and seek" and hiding behind my car



Click on the picture to enlarge340

What must you consider when backing up?

I may not back up on a country road

I may not back up in a residential area

I may not back up if I obstruct or interfere with other traffic

I may not back up on a bus stop

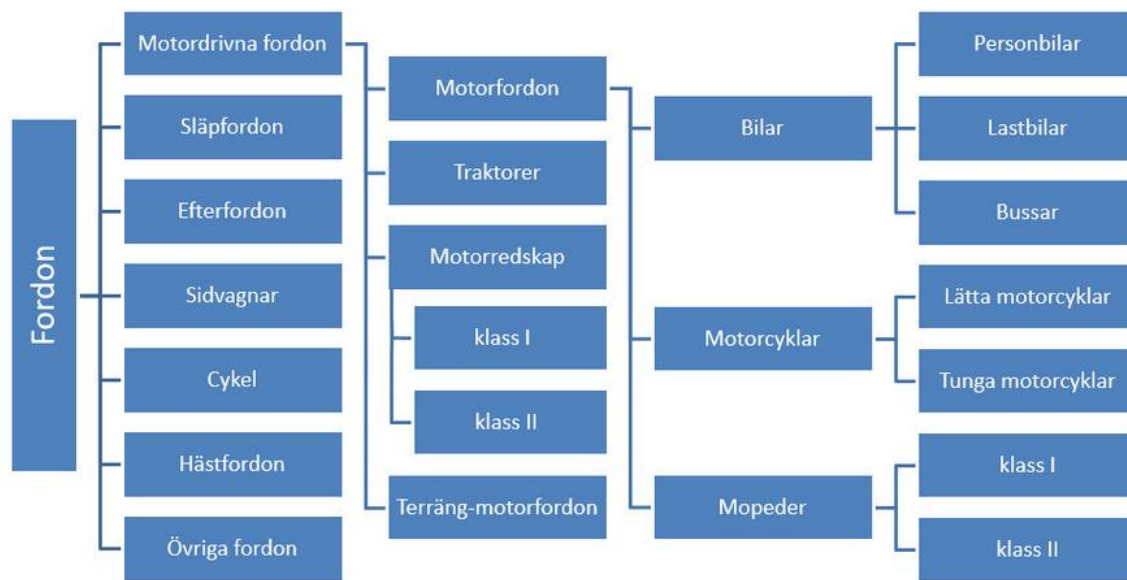
I may not back up in an intersection

Different types of vehicles

There are many different types of vehicles.

Some you may drive with your B license, some others you may not drive. There are also rules and regulations you need to know.

Breakdown into vehicle categories



Click on the picture to enlarge1680

There are a variety of vehicles that can go on the road or in terrain. Different vehicles have different requirements for eligibility, speed limit, vehicle tax and other. It is the Swedish Transport Agency that has created this division in the Act of definitions.

The picture shows, in Swedish, the division of the teams into different vehicle categories and how they are connected:

An example of how to interpret the image:

Vehicles (Fordon) -> Motor vehicles (Motordrivna fordon) -> Motor vehicles (Motorfordon) -> Cars (Bilar) -> Passenger Cars (Personbilar).

Motorized equipment



Click on the picture to enlarge307

There are two classes of motorized equipment.

Motorized equipment class I are designed for a speed **over** 30 km/h.

Motorized equipment class II are designed for a **maximum** speed of 30 km/h

Tractors



Click on the picture to enlarge463

A tractor is a motor-driven vehicle that is type-approved as a tractor. It is a pull vehicle equipped with a coupling device. It must comply with traffic regulations and may be driven on all roads except motorways or expressways.

Tractors are divided into **tractor a** and **tractor b** depending on the speed at which they are designed.

Bikes



Click on the picture to enlarge1721

The bike is a vehicle.

Anyone who rides a bike is considered an unprotected road user.

Anyone leading a bike counts as pedestrian.

Mopeds

Loading image...



Click on the picture to enlarge1722

There are two classes of mopeds.

Class-I moped is a two, three or four wheel motor vehicle designed for a maximum speed of 45 km/h with an internal combustion engine not exceeding 50 cubic centimeters.

If the moped has an electric motor, the power may not exceed 4 kilowatts. If the moped is four-wheel drive and is not powered by gasoline, this may also have a maximum power of 4 kilowatts.

A **drivers license** is required to drive a Class I moped.

Class-II moped is designed for a maximum speed of 25 km/h with a maximum power of 1 kilowatt.

There are also older mopeds that are allowed to drive at 30 km/h. Except for speed, they follow the same rules as a Class II moped.

A **drivers certificate** is required to drive a Class II moped.

Motorcycles

Loading image...



Click on the picture to enlarge1723

A motorcycle is a motor vehicle.

Anyone who drives a motorcycle is seen as an unprotected road user.

Driving a motorcycle requires different skills than those required to drive a passenger car. Therefore, different and also higher demands are placed on a motorcyclist.

Three different classes of motorcycles

A driver with the right age and experience gets the right to take a driving license for larger and heavier motorcycles. The classes available are A1 (light mc), A2 (medium mc) and A (heavy mc).

Cars

Loading image...



Click on the picture to enlarge1724

A passenger car is a vehicle that has a maximum of eight seats in addition to the driver's seat.

There are two classes of passenger cars.

Class I passenger car is a car built primarily for passenger transport.

Class II passenger car (motorhome or truck) is permanently fitted with a body set up as a living space.

Bus

Loading image...



Click on the picture to enlarge86

A car that is established primarily for the carriage of passengers and equipped with more than eight seats in addition to the driver . Buses are divided into light and heavy buses.

Truck

Loading image...



Click on the picture to enlarge266

There are two types of trucks.

A **light truck** has a total weight of **no more** than 3.5 tonnes

A **heavy truck** has a total weight **over** 3.5 tonnes (see picture)

Study Questions

Study questions are answered using the information above including the **Read More** words available. External links are only in-depth.

Loading image...



Click on the picture to enlarge5521

What maximum speed may motorised equipment class II be construct for?

20 km/h

30 km/h

40 km/h

50 km/h

Loading image...



Click on the picture to enlarge5522

Which definition applies to a light truck?

The trucks total weight is up to 6.5 tons

The trucks curb weight is up to 3.5 tonnes

The trucks gross weight is up to 3.5 tonnes

The trucks total weight is up to 3.5 tonnes

The trucks curb weight is up to 6.5 tons

Loading image...



Click on the picture to enlarge5523

The question is about tractors. What is the highest speed you can design a tractor a for?

20 km/h

30 km/h

40 km/h

50 km/h

Cars with internal combustion engine

Apart from a few exceptions, cars with internal combustion engines powered by fossil-based diesel and petrol have been the ones that have completely dominated the car market in general since the car's inception over 100 years ago.

Now a change is underway to drive cars in other ways than with an internal combustion engine that runs on fossil-based fuel. For example with electricity, hydrogen and other types of non-fossil-based fuels.

Fuel system

The car is powered by its fuel in combination with air. A lot of air is consumed. When servicing the car, always check that the air filter is not clogged as this increases fuel consumption.

Fuel consumption

Type of car

How much fuel the car draws depends a lot on the type of car. A so-called SUV (city jeep) consumes much more fuel than a small electric hybrid.

Driving style

Fuel consumption is affected by the speed you keep. The faster you drive, the more fuel it consumes. For every 10 km/h higher speed above 70-75 km/h you drive, fuel consumption increases by 0.5-1.0 deciliters per 10 kilometers.

Fuel consumption is influenced by the cars load

The car's fuel consumption is affected by how much load you have as the weight of the car increases.

If you also have the load on a roof rack or in a roof box, fuel consumption will increase further as the car's air resistance also increases.

Service

Poor maintenance of the car's engine increases the car's fuel consumption. Always service the car according to the owner's manual.

When buying a car

Fuel consumption must be stated for all new passenger cars sold within the EU. Consumption is expressed in liters per 100 kilometers.

Accounting of consumption

The Worldwide Harmonized Light Vehicle Test Procedure (WLTP) is called the driving cycle used to report fuel consumption.

Exhaust system

Cars that have an internal combustion engine also have an exhaust system.

The exhaust system helps to reduce the hazardous exhaust gases. This is done with the help of a catalyst. Diesel cars also have a particle filter. Today, some petrol cars also have particle filters.

The catalyst only works when the working temperature is several hundred degrees. You can use a motor heater to reduce the negative effects of cold starts.

A leaky exhaust system is dangerous as exhaust fumes can find its way into the vehicles interior. Driving with the tailgate open can produce the same effect. When driving in tunnels or in traffic jams, you can also get in exhaust fumes inside the vehicle, you can limit this by turning off the fan or turn on the recirculation mode. See the the owners manual regarding how to do this.

Muffler

The muffler is the part of the exhaust system that reduces noise. If you drive with a faulty muffler you usually bother other road users.

Lubrication system

An internal combustion engine consists of a large number of moving parts that must be lubricated by the car's lubrication system. For this to work, you must have oil in the engine. When doing an external safety check, make sure you have enough oil in the engine. If you have too little oil, the engine can be damaged.

When starting the engine, make sure that the oil pressure is sufficient. With too low oil pressure, the oil does not appear in the lubrication channels present in the engine. If the oil lamp comes on during travel, you must stop the car and engine immediately. Do not start the engine until the fault is rectified.

Electric system

Part of the electrical system has the task of starting the car and ensuring that electrical functions such as lights. This consists of three main parts:

- **Generator** that powered by the motor and generates power to the battery. If the generator is malfunctioning, the battery will be poorly charged and you may have trouble starting the engine.
- **12-volt battery** that stores power. It helps to start the car and power, for example, windshield wipers. If the car does not have a maintenance-free battery, you should regularly check that you have enough battery fluid. This battery is also called a starter battery.
- **Wiring** that distributes power to the various parts that require power, such as electric window lifts. To prevent fire during short circuits, the wiring is protected by fuses. You must know where the fuses are, check in the instruction manual.

Cooling system

Cars with a combustion engine have a cooling system that aims to prevent the engine from overheating. Most internal combustion engines in cars are water-cooled.

As a driver, you must check that there is enough coolant in the cooling system. If coolant is missing, top up with a mixture of water and glycol. The glycol means that the water in the coolant will not freeze to ice when it gets below freezing outside. At a well-equipped gas station there is equipment to check how low a temperature your coolant in the cooling system can handle.

Overheated engine

If the engine temperature indicator is on red, this is a warning that the engine is too hot. You must then stop immediately. You should then check if there is coolant in the cooling system. Feel free to contact your workshop to investigate why there is a lack of coolant.

If you have too little admixture of glycol in the cooling liquid, you can get an ice plug and this can impair the cooling and the engine can overheat.

Warning

Note that you should never open the coolant cap when the engine is hot/overheated. This can cause you to get hot water on yourself.

General systems

The system below is used by all cars, regardless of whether they are powered by an internal combustion- or electric motor.

On the next page you can read about cars with electric motors.

Brake system

With the cars braking system you can stop the car. There are two independent braking systems, a service brake and a parking brake.

The service brake is usually divided into two brake circuits, which means that if one circuit breaks, there is a reduced braking effect with the help of the other circuit. The braking effect is then lower, which gives you longer braking distance.

The braking force is usually transmitted hydraulically, which means that there must be brake fluid in the system for the brake to function.

Steering system

The steering system is used to control the car.

As in most systems, some maintenance is required. A common fault with the steering system is that over time, the joints and gears glitch. You notice the glitch when you can move the steering wheel back and forth without the wheels moving, there should not a glitch.

You may also find that the car "pulls" in some direction. This can occur when the control system is damaged. You may have driven into a curb and damaged the steering system and affected the wheel alignment.

Hydraulic steering servo

Almost all cars today have a steering servo to facilitate steering. You should check regularly that this feature works. If you have a hydraulic steering servo, check that there is power steering fluid in the steering servo. More information can be found in your car's instruction manual.

Electric steering servo

Many cars have an electric steering servo. With an electric steering servo, some cars can even park themselves as it can handle the steering completely by itself.

Towing

When towing a car without steering servo, you can expect the car to be much slower to steer than normal.

Windscreen wiper

Loading image...



Click on the picture to enlarge1194

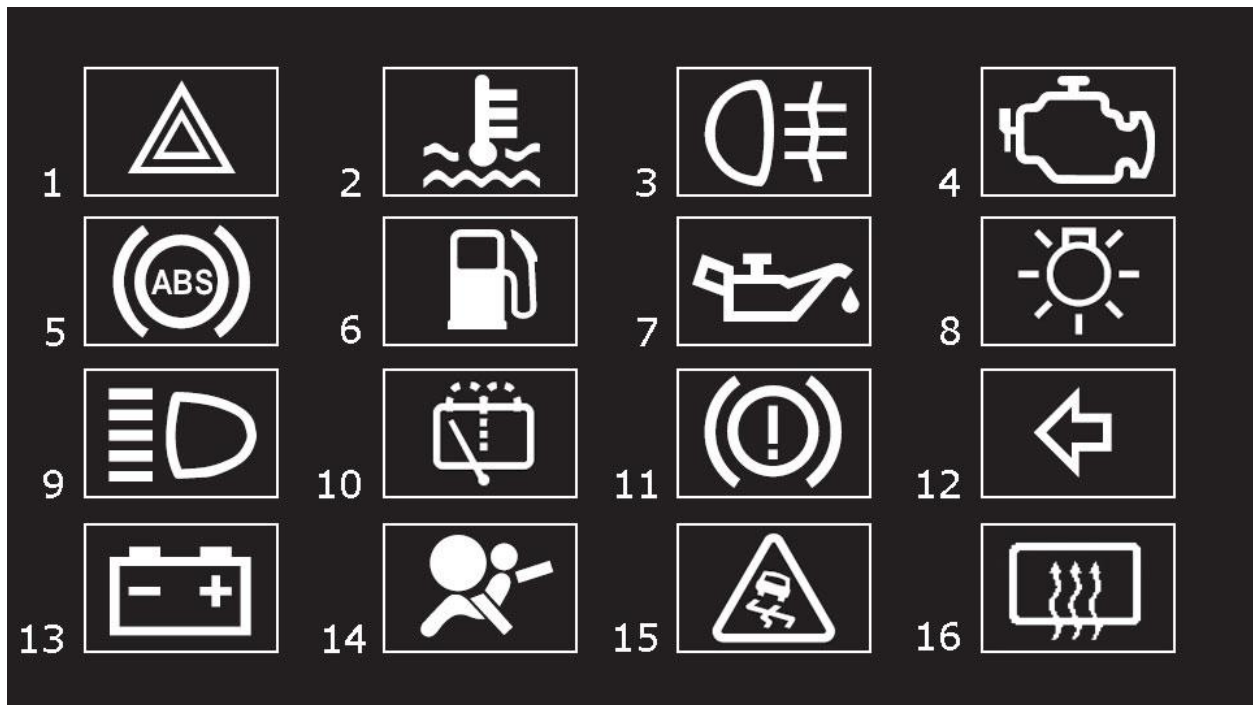
The windshield wiper is used to keep the windshield clear when it is raining and snowing. There is windshield washer fluid to help clean the window. To prevent water from freezing in the windscreen washer, use only windscreen fluid.

Turn off the dryers before switching off the engine

When parking, always turn off the dryers and leave them in their sleep position before turning off the engine. This is so that the wipers do not start automatically when you use the car next time and turn the power back on. This is especially important in winter as the wipers rubber sheet may have frozen on the windshield, which can damage the wipers.

Warning lights

Loading image...



Click on the picture to enlarge ²³¹

When you drive, you should regularly look at the control panel. There are a number of warning lights that normally should be off. The symbols are universal. The picture shows the following common warning lights and its meaning:

1. **Hazard warning lights** - all the turn signals flash simultaneously on the car.
2. **Engine temperature** - the engine is too warm, stop.
3. **Rear fog lights** - rear fog lamp is switched on.
4. **Engine control lamp** - one or more fault codes are stored in the motor control unit.
To find out what caused the lamp to light, the error codes must be read in a garage. A common case is that there is a fault in the exhaust gas purification.
5. **ABS** - something is wrong with the ABS brakes.
6. **Fuel** - it's time to refuel. You probably have a few liters of fuel left.
7. **Oil pressure** - oil pressure is too low, stop.
8. **Lamp broken** - a lamp is broken.
9. **Main beam** - driving on main beam.
10. **Washer fluid** - fill the washer fluid.

11. **Brake Warning** - something is wrong with the brake system, stop. Most commonly, the parking brake is applied or the brake fluid level is too low.
12. **Flasher** - turn signal is switched on (the arrow is usually in both directions).
13. **Charging** - something is wrong with the generator, the battery will not charge.
14. **Airbag** - airbag, something is wrong.
15. **Traction control** - light blinks when the system is "working" and lights up when it is wrong.
16. **The rear window heating** - heating of the rear window is on, consume much power.

Safety check

All systems in a car require some form of maintenance and control. As a driver, you are expected to do what it takes to make the car safe to use.

When you do your driving test for the Swedish Transport Administration, the following is expected:

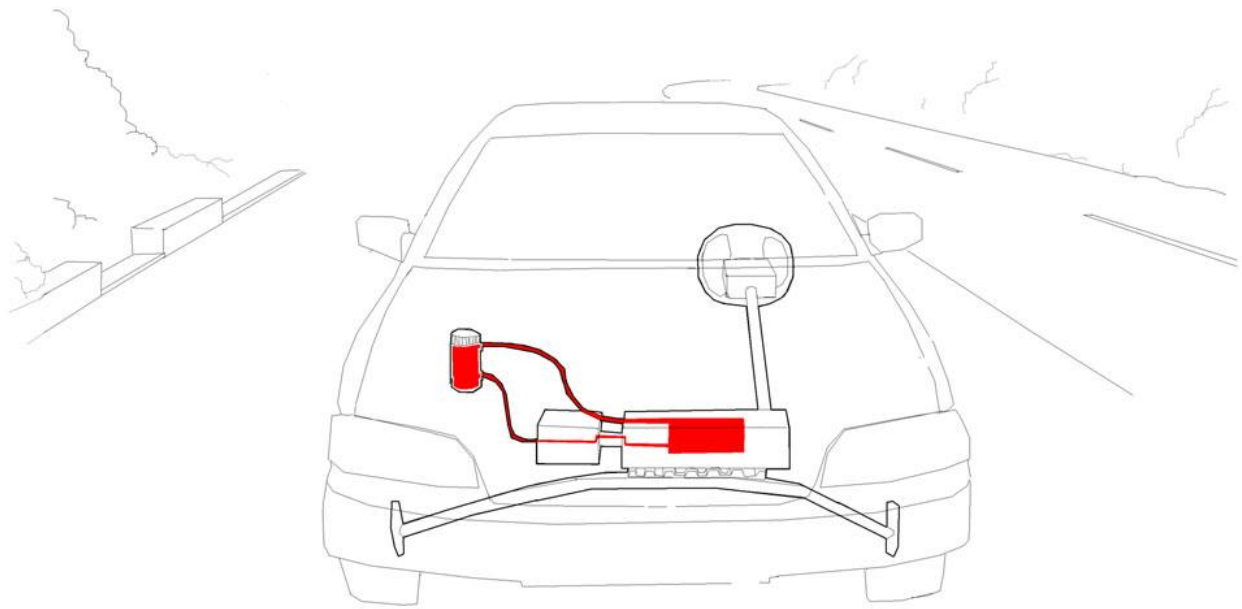
- You should be able to do a safety check and understand what risks can occur if the car does not work properly.
- You should be able to change the controls while driving.

There is a special exercise that goes through how you practically do a safety check.

Study Questions

Study questions are answered using the information above including the **Read More** words available. External links are only in-depth.

Loading image...



Click on the picture to enlarge200

The cars power steering is broken, how does this affect your driving?

The car can be heavy to steer

It is not possible to steer the car

The car can become oversteering

Loading image...



Click on the picture to enlarge229

In the cars service-book it is stated that the cars brake fluid should be replaced every few years, why?

Brake fluid attracts dirt

Brake fluid is viscous and difficult to use

Brake fluid can attract water

Loading image...



Click on the picture to enlarge281

You are on your way home from work. The service brake fails, may you drive on?

No, I have to get a tow truck

Yes, to the workshop

Yes, if I use the hazard lights

Loading image...



Click on the picture to enlarge286

What should you do if the temperature gauge needle rises to the red mark?

Stop and check the coolant reservoir without opening its cap, if any

Stop and check the engine oil level

Drive to a workshop

Provide proper gas so that the fan cools the engine better

Loading image...



Click on the picture to enlarge9741

Your car's battery is discharged and you should, with the help of jumper cables, get power to your battery from another car's charged battery. How to connect the red jumper cable correctly?

I connect the red cable between the positive poles of the batteries

I connect the red cable from the other battery's positive pole to the negative pole of my battery

I connect the red cable from the other battery's plus pole to a body part of my car

I connect the red cable from the other battery's negative pole to the plus pole of my battery

Cars with electric motor

During the 2010s, a change began to take place where electric motors were increasingly used to partially or fully drive cars. Toyota Prius, Nissan Leaf and Tesla Model S were some of the early models.

In 2010, there were less than 1,000 rechargeable electric cars in Sweden and in mid-2025 that number was over 650,000. Most likely, cars with electric motors will increase rapidly and many car manufacturers have already announced their intention to change their focus on the production of cars with internal combustion engines to cars with electric motors.

High Voltage Battery

The car is powered by a special high voltage battery.

[Elbilstatistik, Power Circle AB](#)

Charging System

Electric cars and charging cars with an electric motor is a relatively new phenomenon. The development is also progressing rapidly. It is good to know the basics as you will probably drive a car that needs to be charged with electricity at some point.

Different outlets

A car to be charged has a charging socket and usually one or more charging cables for charging. Some of the most common sockets you may come across are Type 1 (AC) and Type 2 (AC), CCS (DC), CHaDeMo (DC), Tesla (DC), or Schuko (which is a standard grounded household socket). As you can see, you need to read up on what applies to your electric car.

Different charging times

Charging can either be slow (also called destination charging) and occur with alternating current (AC), usually found in ordinary houses. Charging can also be done with fast charging via direct current (DC).

How long it takes to charge mainly depends on how big a battery you have and how much power the charger can provide. On a car with a battery of around 60 kWh, it takes 6-9 hours to charge fully. Fast charging to 80% can be approximately 30-60 minutes.

Risks

Remember that there must be no water or dust in the outlets and sockets. If you start charging with any water or dust, there may be a risk of electric shock or fire. Also, remember to place the charging cable so that no one runs over it, steps on it, trips over it, or otherwise damages it. Always disconnect charging before disconnecting the cable.

Charging sites



Click on the picture to enlarge4334

Charge at home

Most people who have an electric car charge the car mainly at home. There are those who live in their own house and have a charging box or those who live in an apartment building and rent a parking space with a charging box. Those who have their own house can also charge directly via a standard electrical outlet. This is permitted, but you should **always** hire an electrician to check that the electrical system is able to charge via the electrical outlet, as overheating and the risk of fire otherwise exist. A charging box is always safer as it must be installed by an electrician who checks that cables etc., are correctly dimensioned.

Charge publicly

April 12, 2022*, there were over 14,500 public charging stations in Sweden distributed over 2,600 charging stations. The public charging stations can be anything from municipalities and petrol stations to fast-food restaurants and grocery stores. Some cars have built-in systems to find suitable charging points. There are apps for smartphones and websites to help you find them. Some charging stations are free, some take debit cards, some require a subscription, or you have their app or special tiles for their system. There are also charging stations that are locked to a specific car brand.

Find service facilities that offer charging

To find service facilities with charging stations, the location mark Charging Station is sometimes used.

At the charging station, the charging stations are usually signposted with the road sign Charging site which states that only cars with charging may park on the site. There may be time limits. Some charging stations have rules that require charging to be actively ongoing to be able to use the charging station, to increase availability.

Plan

If you have a car with a charge and need to charge during your trip, you should make sure that you know which charging points are available that work for your electric car before you leave home. Find out what rules apply and that you have the systems required to charge. As there are more electric cars, this will probably become easier and easier in the future.

* Source: *Elbilsstatistik.se*

[Elbilstatistik, Power Circle AB](#)

Electricity consumption

Loading image...



Click on the picture to enlarge4330

The amount of electricity needed to power an electric car depends on a number of factors. Some of these are:

- Car brand. Different manufacturers are good at extracting energy from the battery to power the car.
- Driving style. Higher speed and faster accelerations draw more electricity.
- Weather. Consumption increases when it is cold. The use of air conditioning when it is hot also affects electricity consumption.
- Load. If you drive with a lot of load, more electricity is needed.
- Age. A natural degradation of the high voltage battery, depending on the number of years the car was used and driven miles.

Standard consumption

A medium-sized car can use around 1.5 kWh / 10 km, in average consumption according to WLTP. Consumption can also be almost double that for larger cars or cars with other technology.

The energy is stored in the car's high voltage battery

- A chargeable hybrid can often store between 5 - 15 kWh, the range of the car is often between 40-80 km.
- An electric car today can often store between 30 kWh - 100 kWh, the range of the car is often between 200-550 km.

Consumption

The Worldwide Harmonized Light Vehicle Test Procedure (WLTP) is called the driving cycle used to report energy consumption (electricity consumption).

Electric system electric car

There are two types of batteries.

- **A large battery** (battery pack) that is used to power the car and needs to be charged regularly. Its cables are considered high-voltage cables and are orange in order to be clearly visible.
- **A smaller 12-volt battery** that stores power, there are also other volt numbers but 12-volt is most common. The battery helps power windshield wipers, seat heaters and other electrical equipment in the car. The battery has a cable that distributes the current to the various consumers. To prevent fire in the event of short circuits, the wiring is protected by fuses. You need to know where the fuses are, check in the owner's manual.

Discharged 12-volt battery

When an electric car is parked, the large battery is deactivated. To use the large battery, it must be activated. This is done using the 12-volt battery. Therefore, electric drivers may encounter the strange situation that, despite the car being fully charged, you cannot use the car when the 12-volt battery is discharged.

Cooling system electric car

Cars with electric motors

On all types of electric cars, the battery can get too hot but also too cold. Therefore, many electric cars, especially those that only have an electric drive, have technologies for cooling and heating the battery. Read your car's instruction manual if your particular car's solution requires any maintenance regarding these systems to protect the battery pack.

Coolant

The battery pack gets hot as it releases energy. There are air-cooled batteries but on electric car's it is most common with water-cooled batteries. The water must then contain coolant with glycol in order not to freeze when you park the car in the cold. Most often you cannot refill or check the coolant yourself. Read in your car's instruction manual what applies to your particular car.

Regenerative braking

Regenerative braking

Regenerative braking or braking power recovery is a method of braking an electric car by reversing the electric motor. This means turning on the current and a strong resistance in the electric motor is created. On some electric cars this braking effect can be turned off.

With regenerative braking, it is often sufficient to use only the accelerator pedal. When you give gas, the car moves forward, when you release the gas, the car slows down. Keep in mind that the braking effect is affected by how hot the battery is, etc. Prepare yourself before hand so that you won't be surprised!

An advantage of regenerative braking is that the car's high-voltage battery is charged by returning the energy to the battery.

Study Questions

Study questions are answered using the information above including the **Read More** words available. External links are only in-depth.

Loading image...

What does regenerative braking mean?

The car brakes when you give gas

The car brakes when the electric motor in an electric car is reversing when the gas is released

The cars high voltage battery loses power when braking

Loading image...

What abbreviation describes the driving cycle used to report how far an electric car can be driven on its battery?

WLTP

EU

ELFAK

ELKM

Does a battery pack on an electric car need to be cooled?

No

Yes

Owning a car

Owning a car today involves a great deal of responsibility and often a large cost for, among other things, maintenance, insurance, parking and not least the value reduction on the car.

Tip, at the Swedish Consumer Agency there is a special section that is about owning a car. There you can find good advice.

[Köpa bil, Konsumentverket](#)

Registration and inspection

All vehicles in Sweden must be registered and inspected according to special rules.

Registration is made by the person who first puts the car in traffic in the country. This is usually the general agent of the various car brands. You can also import as a private person from another country. Then you must register the vehicle.

Insurance



Click on the picture to enlarge171

As a basis, all motor vehicles must have traffic insurance. The other insurances are voluntary.

We tend to talk about the following insurances :

- Motor third party liability
- Fire, glass and theft insurance
- Vehicle damage insurance

Vehicle tax

Most vehicles in Sweden pay a special vehicle tax. The size of the tax depends on a number of factors.

Below are some factors that may have an impact on the size of the tax:

- Vehicle category
- Tax weight
- Fuel
- Carbon dioxide emissions

- Environmental class

Special tax rules for new vehicles

The tax rules, known as malus, involve an increased vehicle tax during the first three years for gasoline and diesel-powered passenger cars class I and II, light buses and light trucks.

Tip

The rules are complicated to say the least. When you are going to buy a car, you should find out the tax before the purchase using the Swedish Transport Agency's website and the car's registration number. If you buy a new car, it can make a big difference in tax if you choose a model that has high emissions of carbon dioxide.

[Hur bestäms skatten, Transportstyrelsen](#)[Malus - Transportstyrelsen](#)

Vehicle inspection

Loading image...



Click on the picture to enlarge232

Regular inspections is called vehicle inspection (Kontrollbesiktning). At this inspection the vehicle is checked so that it is still safe. You are required to correct the errors found that are seen as to affect road safety. For some errors a new inspection is required.

Passenger cars and lorries not exceeding 3 500 kg in total weight shall be inspected according to the following rules:

- first time no later than 36 months after the month when the vehicle was first taken into traffic
- second time no later than 24 months after the month of the first vehicle inspection
- thereafter no later than 14 months after the month of its previous vehicle inspection.

It is only the month that the vehicle was taken into traffic or inspected which is important, not the date.

Driving ban

If you do not inspect the vehicle on time, it will automatically get a driving ban. The vehicle may only be driven the shortest appropriate route to a repair shop and an inspection company for inspection, provided that the vehicle is in traffic (påställd). If the vehicle is out of traffic (avställd), you may only drive the shortest appropriate route to an inspection company.

Own responsibility

It is the inspection companies, not the Transport Agency, which usually send out the offers for an inspection time, but there is no requirement for them to do so. You are always personally responsible for your vehicle being inspected at the right time, so if you do not get an offer, you must contact an inspection company yourself to reserve time for inspection.

Other vehicles

The inspection rules vary for different types of vehicles. If you own an A tractor, tractor b or any other type of vehicle such as a veteran vehicle, competition motor vehicle or motorcycle, you should check which rules apply to your vehicle at the Transport Agency. It is your responsibility to inspect at the right time to avoid driving bans.

Older legislation

The rules regarding vehicle inspection changed 20 May 2018 to the one described above. Previously, the principle was that the registration number's final figure determined when the vehicle was to be inspected.

[Kontrollbesiktning, Transportstyrelsen](#)

Flying inspection

Police and car inspectors have the right at any time to stop you to control you as a driver and your vehicle. This is called making a "flying inspection ". If you do not have your vehicle in an acceptable condition, you are usually fined and you urgently need to fix the error. You may also be required to demonstrate the vehicle to a workshop or a accredited facility for vehicle inspection.

If it is really serious faults on the vehicle, it can impose a driving ban. It might get salvaged and you may not use the vehicle again until it is repaired and inspected.

Study Questions

Study questions are answered using the information above including the **Read More** words available. External links are only in-depth.

Loading image...



Click on the picture to enlarge190

You drive the green car and have hit the red car. What damages are covered by your car's traffic insurance?

My car's damage

All personal injuries and damage to the other party's vehicle

Damage to my car in connection with theft



Click on the picture to enlarge5798

If you do not inspect your vehicle in time, what will happen?

The car will automatically get a driving ban

I get a reminder

I can drive two months after my inspection period

I contact the Vehicle inspection and request a deferral



Click on the picture to enlarge5800

You have bought a six-year-old used car. How often must this inspection be carried out?

No later than 14 months after the month in which its previous inspection was carried out

No later than 24 months after the month in which its previous inspection was carried out

No later than 36 months after the month in which its previous inspection was carried out

No later than 48 months after the month in which its previous inspection was carried out

Who is responsible for inspecting the vehicle on time?

It is the owner of the vehicle who is responsible for inspecting the vehicle on time

The Swedish Transport Agency is responsible for ensuring that all vehicles are safe, so they send out a notice for inspection

Do all types of vehicles have the same rules for regular vehicle inspection?

Yes, all types of vehicles must be inspected the first time within 36 months, the second time within 24 months and then within 14 months

No, the rules for inspections vary for different types of vehicles

Tires

There are different types of tires. Common is that the condition and quality of the tires greatly affect your road grip.

It is particularly important that you have good tires when there are wet roads or winter road conditions.

Types of tires

Tires are usually divided into two groups:

- Summer wheels
- Winter tires

Energy labeling

When you are going to buy new tires, you should take into account the tires' energy label in order to have as little environmental impact as possible. It is also on this marking that you can see if the tire is approved as a winter tyre.

Margins of safety

Regardless of the type of tire you choose, the primary thing is always to keep a good distance and not drive faster than you have time to stop if something happens.

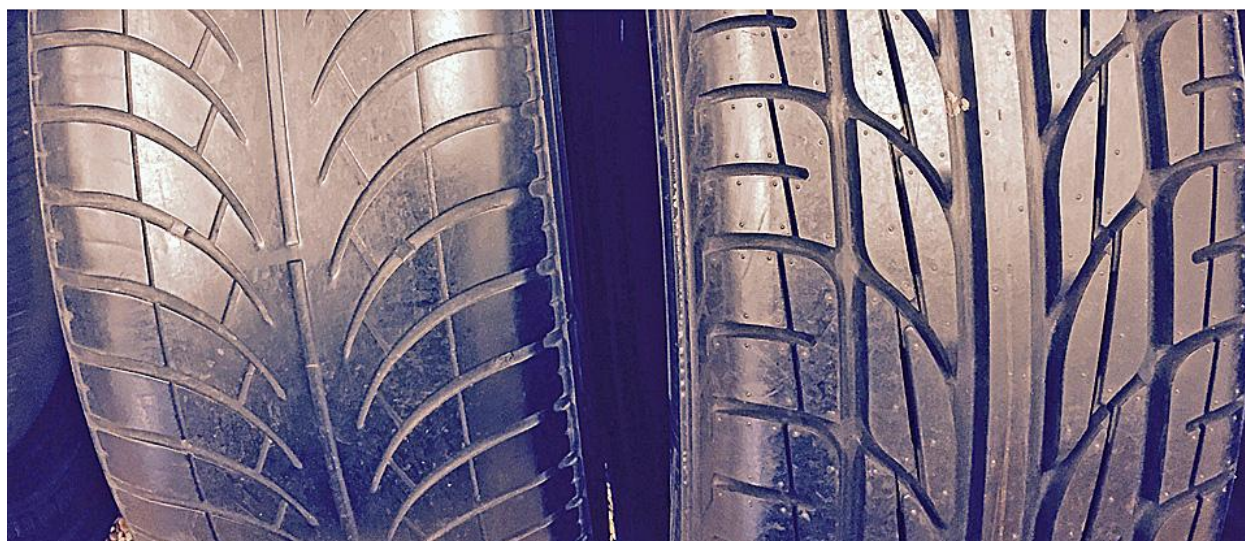
Tip, **never use** winter tires to drive faster in winter. Don't be lulled into a false sense of security. Use the increased safety the winter tire provides in winter road conditions to increase your safety margins!

Summer tires



Äldre godkänt däck

Nytt däck



Click on the picture to enlarge1856

A new summer tire has approximately 7-8 mm tread depth. The pattern on a summer tire is often fine to give a good grip on dry and fine roads.

The patterns most important function is to lead away water when it rains so that the tire does not lose contact with the road surface. It is easy to see that a new tire with more tread depth can handle more water than an older, more worn tire , even if the older one is approved. If the tire does not succeed in leading water away, you get something called hydroplaning and then lose all traction.

A summer tire is generally made of a rubber that can withstand heat but that hardens in lower temperatures. Even at five degrees, the tires characteristics has deteriorated so much that they work very poorly in winter conditions.

A summer tire must have a **minimum of 1.6 mm tread depth**.

Winter tire



Click on the picture to enlarge1857

Requirements for vehicles with a total weight up to 3500kg

Winter tires used on cars and on trailers for cars with a total weight of no more than 3,500 kilograms must:

- be marked with the alpine peak/snowflake symbol

or

- be dubbed.

A new winter tire has about 9 - 10 millimeters of tread depth. A winter tire's pattern is often rough-cut to provide good grip on roads with snow. **The tread depth for winter tires used in winter road conditions must be at least 3 millimeters.**

A winter tyre, compared to a summer tyre, is usually made of a soft rubber compound that withstands heat less well but is better able to maintain its softness in lower temperatures. This applies above all when the winter tires are new.

If the car has studded tires, the trailer must also have studded tires in winter road conditions.

Previous labeling

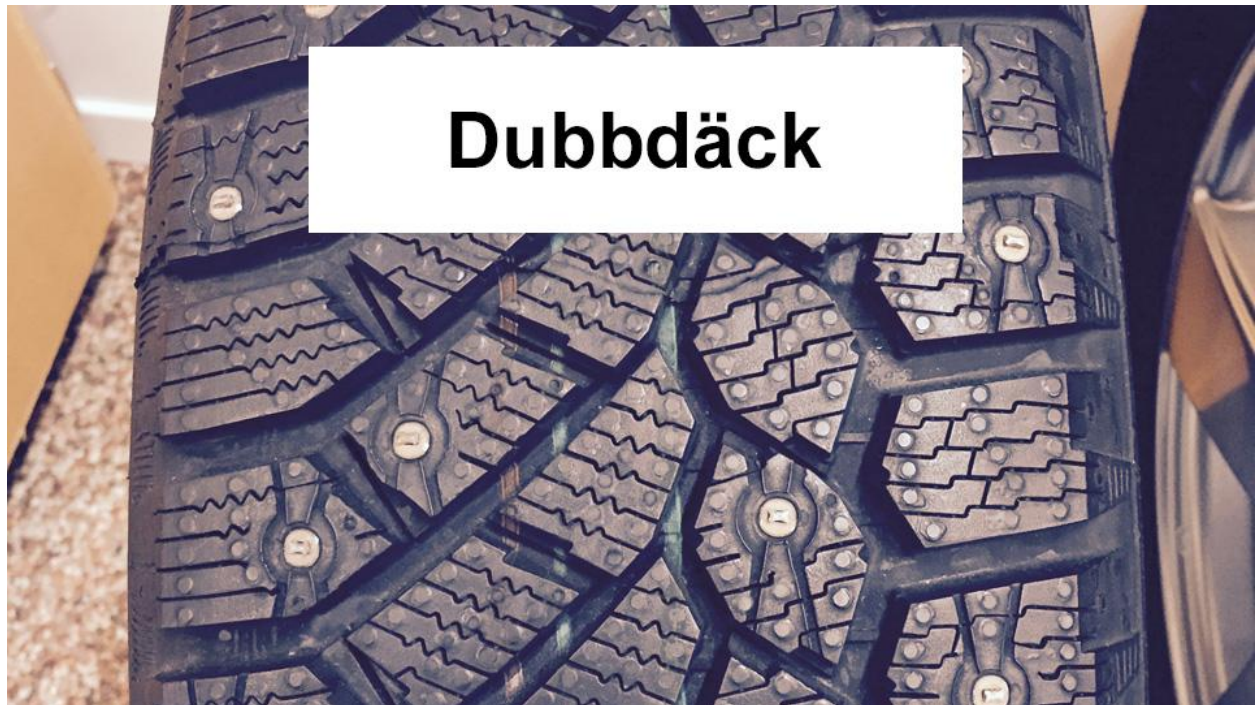
Until November 30, 2024, tires marked only with "M+S" (Mud and Snow) were considered approved winter tires for passenger cars with a total weight of up to 3,500 kg.

Exception

The law does not apply to mopeds and motorcycles.

Studded tires prohibited

Loading image...



Click on the picture to enlarge¹²¹

A studded tire is a tire with a rough pattern, fitted with spikes sticking out a little bit from the tire itself. These small metal spikes allow the tire to have a better grip on ice than the tires without spikes. However, you often have longer braking distances with studded tires on dry asphalt.

Once you have studded tires on your car, you must have it on all four wheels on the car.

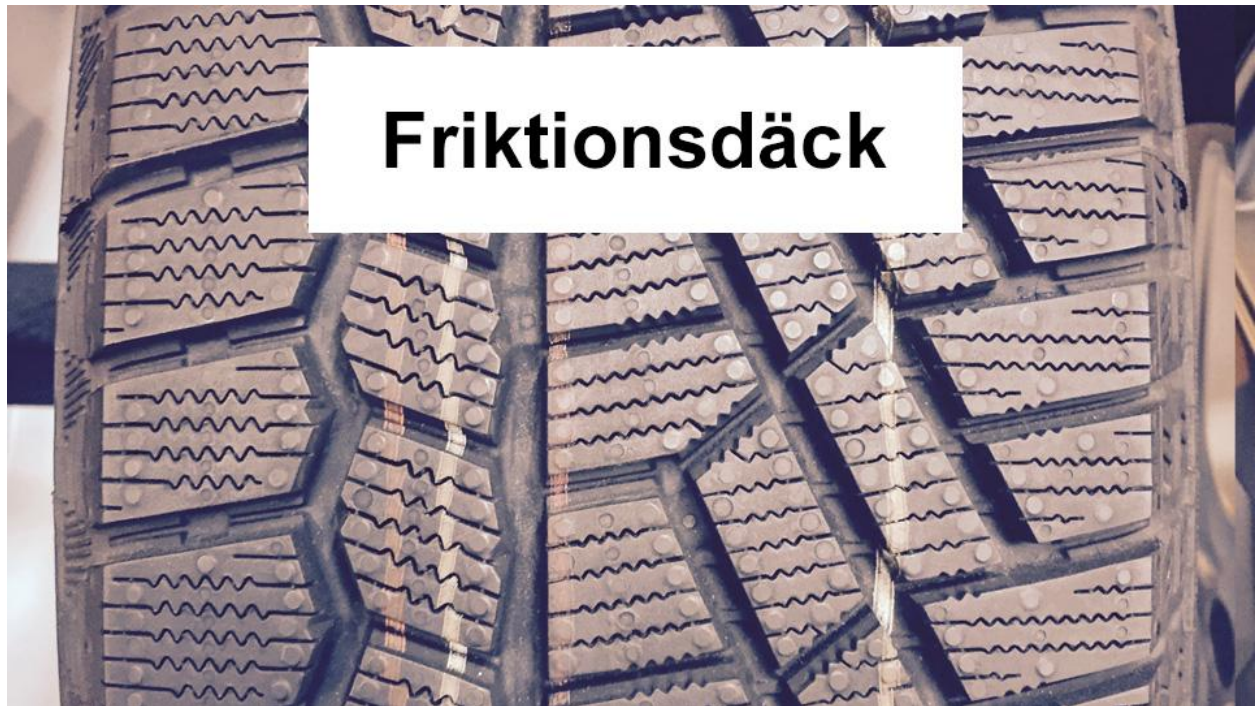
Trailer

If you drive a trailer in winter conditions this must also have studded tires. If you drive a car without studded tires the trailer can have studded or non-studded snow tires.

[Vinterdäck, Transportstyrelsen](#)

Friction tires

Loading image...



Click on the picture to enlarge156

Friction tires are studless winter tires made of a very soft rubber mixture. These tires work very well on snow and slush.

Studded tires are much better on ice than friction tires. Where you live in Sweden often determines what type of tires you choose.

All-year tires

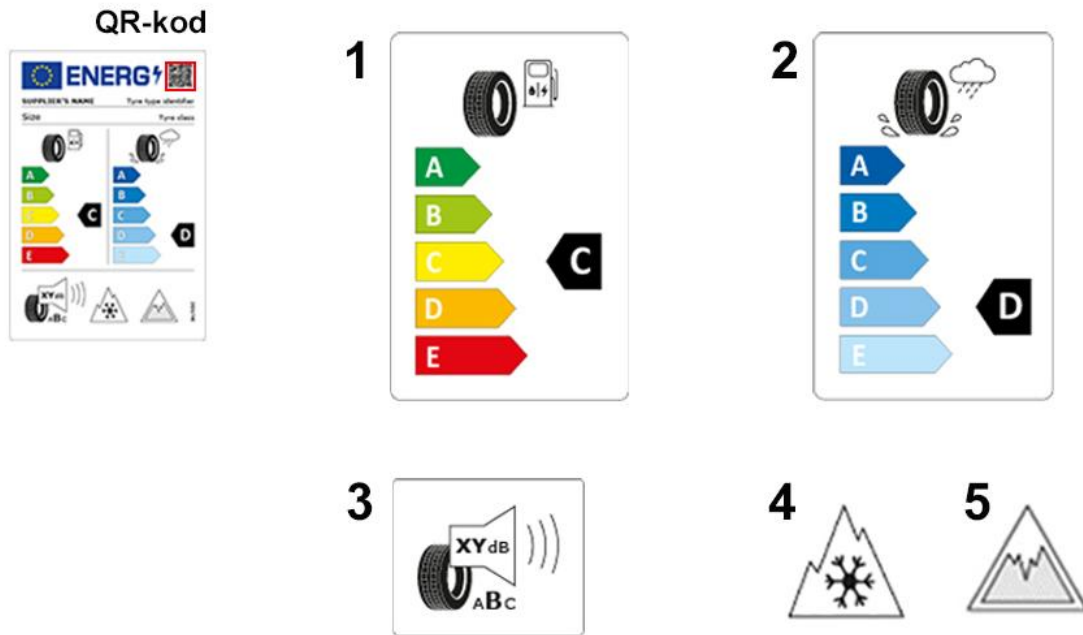
This type of tire can handle dry and wet roads, a bit of slush, and cold weather, but they cannot keep the car's driving characteristics at as safe a level, under actual winter conditions, as real winter tires. However, they are marked with the marking required of winter tires, so you comply with the winter tire law.

An all-year tire is a little worse than a summer tire in the summer, while it is a little worse than a winter tire in the winter when there are winter road conditions. The tire is a kind of compromise.

It is legal to drive with these all year round, i.e., both in summer and winter. What you choose naturally depends on where you live and which road laws apply where you drive most.

Energy labeling for tires

Loading image...



Click on the picture to enlarge [3147](#)

This energy label for tires came into force on 1 May 2021. You recognize this marking on the QR code at the top right of the energy label and the buyer should be able to scan this for more information.

The scale for energy efficiency and grip is A to E.

If the energy labeling label has snow grip labeling, the tire is an approved winter tire according to law. With this label, it will be easier to choose winter tires than with older labels.

For a while, there will be tires for sale that have this energy label with the scale A-E and other tires that have the energy label with the scale A-G.

Great savings potential with efficient tires

Tires account for 20 percent of the vehicles' fuel consumption and the most energy-efficient tires are estimated to reduce consumption by up to 10 percent. At the EU level, energy labeling is expected to save 5% of the total use of fuel for transport in the long run. Lower fuel consumption contributes to reduced carbon dioxide emissions.

Description of the symbols (see picture)

1. **Rolling resistance** - How easily does the tire roll? This is the same as fuel economy and is graded from A to E where A is most efficient. The grading includes a color scale from green to red.
2. **Wet properties** - What grip does the tire have when braking on wet asphalt? Graded from A to E where A is the best grip. The grading includes a color scale from dark blue to light blue.
3. **Ambient noise** - How much does the tire sound outside the car? The result is reported both with the number of decibels and with the scale A to C where A is quietest.
4. **Snow grip marking** - Indicates that the tire is approved by law as a winter tire. The brand is also called alpine top/snowflake.
5. **Ice grip marking** - The tire is, according to the manufacturer, adapted for icy roads in Nordic countries. Mandatory and common labeling for EU countries

Energy labeling is mandatory and common to EU countries

The market control authority in Sweden is the Swedish Energy Agency.

Examples of tires that are not covered by the rules on energy labeling

- Studded tires
- Retreaded tires
- Spare tires for temporary use
- Used tires

Tire tread depth checker

Loading image...



Click on the picture to enlarge313

To measure the tread depth on a tire, you can use special treads from a car accessory company.

Make sure you measure the most worn part of the tire.

Pattern depth summer tires

- 1.6 millimeter is the law's minimum requirement
- To increase safety, it is recommended to have a tread depth of at least 3.0 millimeter

Tread depth winter tires

- 3.0 millimeter is the law's minimum requirement when the winter tire law applies (even during this period, 1.6 millimeter is sufficient if it is not winter road law)
- To increase safety, it is recommended to have a tread depth of at least 5.0 millimeter

The "best" tires

Use the least worn tires at the rear

A common type of accident is that you skid with the rear wheels and, in the worst case, lose control of the vehicle. Therefore, you should always have the least worn tires, the best tires, at the rear.

If you do not replace all tires, do this

If you have four used tires but only want to replace two of them, you should place the used tires at the front and the two new ones at the rear.

Front or rear wheel drive

You should always install the best tires in the rear, regardless if you have a front or rear-wheel drive.

Study Questions

Study questions are answered using the information above including the **Read More** words available. External links are only in-depth.



Click on the picture to enlarge111

Your tires are worn differently but still legal. Where do you mount your least worn tires?

It does not matter how I place the tires

I place the least worn tires in front

I place the least worn tires in the rear

During the period 1 December - 31 March, the winter tire law applies. When it is February 1st, what is the minimum tread depth allowed if it is not winter road conditions?

1,0 mm

1,6 mm

3,0 mm

3,6 mm

Loading image...



Click on the picture to enlarge233

Normally the car go straight when you are holding the steering wheel loosely, but now you notice it "pulls to the right", what can create this?

I have too little air in the right front tire

I have too little air in the left front tire

My tires are worn different



Click on the picture to enlarge285

What is correct about the tires on the car?

Worn tires increases the risk of aquaplaning

High speed reduces wear on the tires

Low speed increases wear on the tires

The pattern depth must be at least 1.0 mm if it is a summer road

Why is 7 mm tread depth better than 3 mm?

It gives better traction on a dry asphalt road

It gives better traction on a wet asphalt road

It gives better traction on a paved road, whether the road surface is dry or wet

Vehicles, different weights

The weights of a vehicle are described in many different ways.

Total weight

Total weight is the weight obtained through adding the curb weight and the vehicle's maximum load.

Example:

Curb weight 1 500 kg + Max load 500 kg = Total weight 2 000 kg

Curb weight

Curb weight is the combined weight of the car with the driver when it is unloaded and fully equipped. This means that the car has fuel, engine oil, water, tools and spare wheels that belong to the vehicle. Regarding the Curb weight, the driver's weight is counted as 75 kg.

The operating weight is stated in the car's registration certificate and can vary from car to car. The curb weight is important to know because it affects which trailers you can tow with the car.

Maximum load

The car's maximum load (Maxlast) is written in the car's registration certificate and is the difference between the vehicle's total weight (Totalvikt) and kerb weight (Tjänstevikt). This must not be exceeded.

Driver and car

The weight of the driver is not included in the maximum load but each passenger counts as 70 kg. This means that you seldom can load as much as you think when you have four passengers.

An example; a car which has a maximum load of 380 kg can carry 100 kg if the four occupants weigh exactly 280 kg.

Gross weight

Gross weight means the vehicle's current weight at a given time. It is this weight that is affected when loading your vehicle.

Think like this:

- Gross weight = actual weight at weighing
- Total weight = theoretical value according to registration certificate

Gross weight is not found in the vehicle's registration certificate.

Drive with a trailer

The registration certificate states how much a towing device is allowed to pull for an unbraked and a braked trailer, respectively. The weight referred to is the gross weight of the trailer. This can sometimes limit how much of the trailer's load capacity you can use.

Restriction for vehicle combination

There may be a maximum permissible gross weight for a given vehicle combination. This must never be exceeded, even if the car and the trailer separately may weigh more.

Axel load (concept)

Loading image...



Axeltryck = 1 axel

Click on the picture to enlarge⁹²⁷

Swedish legislation:

The total static weight that the wheels on a wheel axle transfer to the roads surface.

Boogie pressure (concept)



Boggitryck = 2 axlar

[Click on the picture to enlarge931](#)

Swedish legislation:

The total static weight that the wheels in a boogie carry over to the roadway

Triple shaft pressure (concept)



Trippelaxeltryck = 3 axlar

[Click on the picture to enlarge985](#)

The total static weight that the wheels in a triple axle transfer to the roadway.

Study questions are answered using the information above including the **Read More** words available. External links are only in-depth.

[illegible]

Which of the following describes a vehicles total weight?

Curb weight plus gross weight

Curb weight plus maximum load

Gross weight plus maximum load

Which of the following concepts describe the weight that increases when you load your car?

Gross weight

Curb weight

Total weight

Maximum load

The question is about the cars curb weight. Which statement is correct?

Only the car is included in the cars curb weight

The cars curb weight is when the car is ready to drive including max load, fuel and driver

The curb weight of the car is when the car is not loaded in drivable condition with fuel and driver

The curb weight is the weight of the car plus its maximum load

Where can you find information about how much the car can load?

The registration certificate

The vehicle inspection instrument

The insurance card

Vehicle load

Important knowledge about the load:

- As a rule, everything that does not belong to the vehicle itself is counted as load (that which is not included in the service weight)
- You may not load so you obstruct the drivers view.
- You may not load so you obscure any lights or registration plates on the car.
- The load must be secured so that it can not cause any damage.
- If you load outside the vehicle and use straps, remember to retighten them after driving for a while.
- Be careful how much you load on a so-called roof rack or in a ski box. Too heavy loads high up can make the car wobbly.
- You may not have more people in the car than it is registred for, see the car's registration.
- The cars maximum load must not be exceeded, see the vehicle registrations. The driver is not counted since he or she is included in the cars curb weight, but each passenger counts as "load".

- Try to get heavy loads as low and as far forward as possible to reduce the risk of the "rear end" being pushed down and increasing the risk of oversteer.
- When the rear end is pushed down, others can be dazzled by your dipped beam shining higher and may hit oncoming drivers right in the eyes.
- Do not drive with the tailgate open, you risk getting exhaust into the car.

Controllability and stability

When you drive your own car, you are used to how it usually behaves in different situations. For example, you know how fast you can take a curve without the risk of skidding and other behaviors that are dangerous.

When you borrow or rent a car, you are entirely unaware of how the car will behave. To reduce the risk of unexpected situations, you can drive with extra attention until you learn how the car in question behaves.

Changes

The most significant risk is when there are sudden changes. Imagine that you are going on vacation and you load the car entirely in the trunk while you are leaving with four people. The handling and stability of the car have definitely been affected by what you have loaded and when you drive the car may turn a lot more than you expect. This is usually called because the car has been oversteered.

Length

A motor-driven vehicle with attached trailers must not be longer than 24 meters including protruding load. This is the basic rule.

The Swedish Transport Administration or a municipality that is a road manager may, however, stipulate that on a certain road, part of road or stretch of road, a vehicle train may be driven with a length of no more than 34.5 metres, including the load. Please note that there are a large number of regulations that vehicles and drivers of these long vehicle trains must comply with.

Projecting load

When loading a vehicle, any load that projects forward or more than one meter backwards must be clearly marked. The marking must be done with a flag or equivalent.

Exception when load protrudes forward

If the load protrudes no more than one meter forward, it does not need to be marked if it is clearly visible to other road users. Imagine driving a bed on the roof that juts out a meter forward. This is clearly visible and does not need to be marked out. If, on the other hand, you drive a narrow box, it is a danger and must be marked out.

Darkness

During darkness, the marking must be done with light and reflex. Use white forward and red backward.

Road signs

There are road signs that limit vehicle length in certain locations. It happens that long vehicles are restricted inside urban areas. This is regulated in the local traffic regulations, which above all professional drivers must take into account.

Width

Loading image...



Click on the picture to enlarge72

A vehicle may, including load, not exceed 260 cm wide (Bus 255 cm). This applies when driving on public roads. If you drive on private roads other rules may apply.

If you need to load something that is wider than your vehicle it is permitted if the load does not stick out more than twenty (20) centimeters on either side, and the total width does not exceed 260 cm (Bus 255 cm).

If you need to drive something that is wider than 260 cm (Bus 255 cm), you must seek permission. It is the Swedish Transport Administration's regions or the municipalities that examine these applications.

Exception

- The width, including the load, on a light motorcycle must not exceed 120 cm.
- Motorized equipment and agricultural machinery may be wider than 260 cm without special permission, see the picture. In harvesting periods it is not uncommon to meet a combine which is 400 cm wide, in other words 4 meters wide.
- Unpacked hay or the like can stick out more than 20 cm sideways. The maximum width of the vehicle in question may never be exceeded.

[Största tillåtna längd och bredd, Transportstyrelsen](#)

Height

Loading image...



Click on the picture to enlarge550

Generally speaking, a vehicle may be at any height. However, it is very unusual that vehicles are higher than 4.5 meters as this is the so-called free height when passing tunnels and viaducts. If the free height is lower, this must be indicated on a road signs in connection with the passage.

Marking load

Provided that you do not exceed the maximum length (24 meters), the load may stick out in the front and in the back.

The following limits apply to the marking of load. It is the outermost part of the load that must be marked:

- Load that sticks out **in the back** must be marked if it sticks out more than one meter behind the car. Flag or equivalent in daylight, in the dark **red** lantern and **red** reflector applies.
- Load that sticks out **in the front**, in front of the car, must always be marked, with one exception. If the load sticks out less than one meter and is clearly visible to other road users, it does not need to be marked. Flag or equivalent in daylight, in the dark **white** lantern and **white** reflector applies.

What is "clearly visible"?

Imagine that you are driving a bed on the roof that sticks out one meter in the front. This is clearly visible and does not need to be marked. If, on the other hand, you drive a narrow bar, it is a danger and must be marked.

Towing

When towing another vehicle and the tow-rope, or rod, is more than two meters, you need to mark this clearly with a flag so that no one drives between you and the one you are towing. You may not tow at a higher speed than 30 km/h.

Hazard lights

You may use hazard lights when towing, but only on the vehicle you are towing, never on the towing vehicle.

Darkness

If the vehicle you are towing lacks lights, and it is dark outside, you have to mark it with red light and a red reflector in the rear. Put the reflector as low as possible.

Hard shoulder

If there is a hard shoulder, you must use this.

Motorway or expressway

There is a general ban on towing on motorways and expressways.

If you are forced to make an emergency stop on a motorway / expressway, you should hire a professional tow truck. If you see someone standing along the edge, do not stop as there are great risks to stop on these highways.

Motorcycle

A two-wheel motorcycle may never tow or be towed by another vehicle. You also may not, when you drive a two-wheeled motorcycle, pull or push any other road user in front of you.

Bicycle and moped

On a bicycle and moped, you may not let another vehicle pull or tow you. You also may not pull another vehicle or push it in front of you with a moped.

The vehicle manufacturer may not permit towing

Some cars with electric motors must not be towed, according to the manufacturer. Check your car's instruction manual what applies to your car.

Trailer

Your driving license authorization affects which trailers you may drive with a car or light truck.

Trailer with driving license B (passenger car / light truck)

You may always drive a car with a trailer that has a maximum total weight of 750 kg.

You may also drive a car with a trailer that has a total weight that exceeds 750 kg if the combined total weight of the vehicle combination is not higher than 3,500 kg.

Trailer with driving license B 96 (extended B)

You may drive a car with a trailer where the total weight of the vehicle combination exceeds 3,500 kg but is not higher than 4,250 kg.

Trailer with driving license BE

You may drive a car with one or more trailers where the total weight of the vehicle combination exceeds 4,250 kg if the total weight of the car is at most 3,500 kg and the total weight of the trailer or trailers is at most 3,500 kg.

Vehicle terms:

- The trailers gross weight (Bruttovikt) (curb weight (Tjänstevikt) + current load) must not exceed the cars maximum weight for an unbraked or braked trailer.
- The total combined vehicle weight may not be exceeded. The total combined vehicle weight can be found in the car's registration certificate.

Speed limit terms trailer without brakes

Speed rules are a bit special when it comes to a trailer without brakes. The trailers curb weight **or** total weight determines the maximum speed. If the trailer is unloaded the curb weight apply, if you load anything (no matter how little), the total weight apply. The weight one compare to is the cars curb weight.

Prerequisites for the example below : The cars curb weight is 1000 kg, trailers curb weight is 300 kg and the trailer total weight is 750 kg.

- **Unloaded trailer:** You may drive at maximum of **80** km/h because the trailers curb weight of 300 kg does not exceed half the car's curb weight.
- **Loaded trailer:** You may drive at maximum of **40** km/h because the trailer total weight of 750 kg exceed half the cars curb weight.

Speed limit terms trailer with brakes

- If the trailer has brakes, it may drive at maximum of **80** km/h. This limitation also applies to the motorway and expressway.

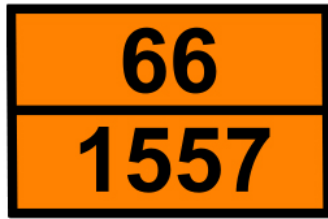
Lighting terms:

- The trailer must be equipped with taillights, brake lights, turn signals and license plate lamp.
- The trailer must be equipped with rear red triangular reflectors. Orange reflectors on the side and white on the front.
- If the trailer has a total weight of 3500 kg it should be equipped with a special EU-reflector.

[Hur tungt släp får du köra med B-behörighet, Transportstyrelsen](#)

Dangerous goods

Loading image...



**Tankfordon eller andra fordon
med särskilda behållare**



**Fordon som transporterar
farligt gods som styckegods**

Click on the picture to enlarge 140

Everyone must know the basic rules for dangerous goods that are marked with orange warning signs both front and rear.

There are two types of warning signs.

- Signs with numbers in two rows attached to tanker vehicles or other vehicles where there are special containers for the goods.
- Signs without numbers attached to the vehicles transporting cargo.

In addition to the above warning signs, the goods should be marked with warning labels.

Action in case of an accident

If you arrive at an accident and you see that a car involved has an orange warning signs you should act in a different way than in an ordinary accident. Since you can not possibly know what kind of substance that is transported, you must wait for rescue personnel with protective equipment and special training for such situations.

What you can and should do is:

- Call 112 and alert them.
- Inform the emergency center that you have seen the orange signs. Tell them what it says on the signs if you can see this.

- Put your car so that it is clearly visible, turn on the hazard lights.
- Put up the warning triangle.
- Do not smoke.
- Do not stand downwind from the accident scene since there might be toxic gases involved.

Dangerous goods for private persons

Loading image...



Click on the picture to enlarge4587

Special demands

As a basic rule, only those with a permit, special training, and specially marked vehicles may transport dangerous goods.

Exception

For practical reasons, there are exceptions for private individuals:

- Private individuals may transport dangerous goods in consumer packaging that are sold in the retail trade and that are intended for their own use.

- You may transport dangerous goods, which is a natural part of what you transport. For example, the fuel that you have in a boat when it is on a trailer behind the car.

Restrictions

- When the dangerous goods consist of flammable liquids (e.g. petrol and diesel) which are transported in refillable containers and filled by or for private persons, then the amount per container must not exceed 60 liters. The total amount of flammable liquids must not exceed 240 liters on a vehicle.
- Private individuals may transport hazardous goods that are waste to recycling centers and the amount may not exceed 30 kg per vehicle. For lithium batteries, a maximum of 5 kg net weight applies. It may also be good to check the recycling center's rules.
- More significant types of packaging such as IBC containers, large packaging, or tanks filled with dangerous goods must not be used.

Checklist weights trailers

Checklist to understand the rules around cars and trailers.

Towing device

How heavy a trailer can the vehicle tow with the towing device? In this case, it is always the **gross weight** that counts.

To know the answer, you need to find out if the trailer is unbraked or braked.

Then you must check the registration certificate what is the maximum weight for unbraked and braked trailers.

The rule may mean that you cannot use the maximum load of a trailer as it is in relation to the towing device that the gross weight is counted.

Combined total weight

What driving license authorization is required to tow a certain trailer? In this case, it is always the **total weight** that counts. It does not matter if the trailer is loaded or not.

If the trailer's total weight **does not exceed 750 kg**, you can always drive this trailer as it counts as a light trailer. When the towing vehicle is allowed to weigh a maximum of 3 500 kg, the combined total weight in this case with a light trailer can be 4 250 kg.

If the total weight of the trailer is **higher than 750 kg**, the following applies as the trailer is counted as a heavy trailer:

- if the combined total weight is not more than 3 500 kg, a driving license for B is sufficient,
- if the combined total weight is not more than 4 250 kg, a driving license for B extended (B96) is required,
- if the combined total weight is at most 7 000 kg, a driving license for BE is required.

Registration certificate

The towing vehicle's registration certificate states the maximum total weight of the trailer for a B driving license.

Study Questions

Study questions are answered using the information above including the **Read More** words available. External links are only in-depth.

Bil		Släpvagn, bromsat	
Tjänstevikt	1 320 kg	Tjänstevikt	400 kg
Maxlast	600 kg	Maxlast	1 100 kg
Totalvikt	1 920 kg	Totalvikt	1 500 kg
Högsta vikt obromsat släp	500 kg		
Högsta vikt bromsat släp	1 400 kg		

Click on the picture to enlarge1840

Look at the picture. Can you use the maximum load on the trailer with this vehicle combination?

Yes, since the trailers curb weight is only 400 kg there are no limitations

No, since the combined total weight is less than 3 500 kg

No, I can only load 1,000 kg on the trailer as otherwise the gross weight will be too high

Loading image...



Click on the picture to enlarge132

Which option is correct when you attach a trailer to your car?

I get better grip on the rear wheels if I have a low towbar noseweight

I can experience the car as winding if I have a high towbar noseweight

I get better driving characteristics if I load the trailer a lot

I get worse handling and increased wear on my rear tires if I have high towbar noseweight

Which statement is correct?

I am not allowed to use low beam

I must not use hazard warning lights on the vehicle I am towing

I may not tow a car with the brake servo out of order

I may not tow a two-wheel motorcycle

You should take lithium batteries that you have used for your lawnmower to a recycling station. What is the maximum weight of the batteries so that it is not considered a transport of dangerous goods?

The net weight may be a maximum of 1 kg

The net weight may be a maximum of 5 kg

The net weight may be a maximum of 15 kg

Loading image...

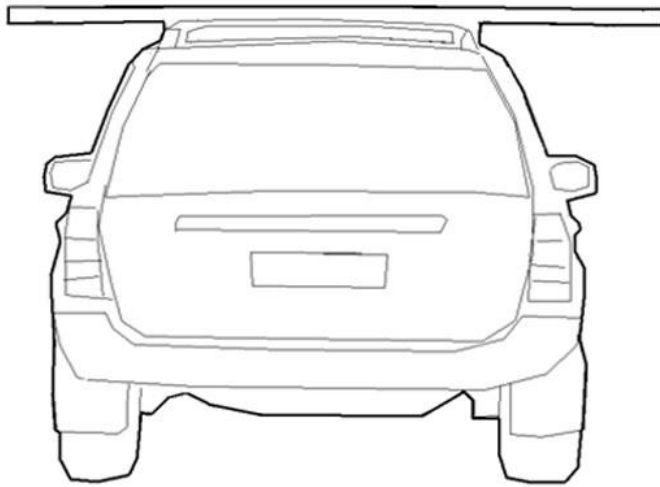


Click on the picture to enlarge137

What characterizes understeering?

The car is turning less than I turn the steering wheel

The car is turning more than I turn the steering wheel



[Click on the picture to enlarge](#)131

What is correct regarding how much cargo gets to shoot out sideways?

The load may project no more than 20 cm on each side as long as the overall width is not exceeding 260 cm

The load may project 30 cm on one side and 10 cm on the other hand, as long as the width of the vehicle does not increase more than 40 cm

The load may project more than 20 cm if it is marked with pennants

The load may project no more than 20 cm on each side as long as the overall width is not exceeding 250 cm

Loading image...



Click on the picture to enlarge344

What should you consider when you load your car?

I must always place any cargo in the trunk

I may not place any load so that my view ahead or to the sides are obscured

I do not think about anything special since the car is made to drive with load

I get a shorter braking distance when I unload the car because it then becomes heavier

Concepts and definitions

Traffic is governed by laws and regulations.

In order for the regulations to work well, it is important that everyone being a part of the traffic system has a common idea of what the written word means. Therefore, there is a law and regulation that governs all concepts and definitions, eg what is a bus?

- The Road Traffic Definitions Act (law) (2001: 559) describes many of the concepts that exist, for example Motorcycles, Off-road vehicles, Horse vehicles. The Act (Law) is determined by Parliament (Riksdagen).

- Regulation (2001: 651) on road traffic definitions describes many of the definitions that exist, eg Traveling route, Pedestrian street, Terrain. The regulation is determined by the government (Regeringen).

Offence

When you break a traffic rule, there are different types of penalties. Some traffic violations are considered misdemeanors and you can accept a penalty with a fine instead of taking the matter to court.

An example is a minor speeding violation which is considered a misdemeanor and therefore you can accept a fine on the spot.

Day fines

Day fines are a form of fine where the amount is determined by the convicted person's income. The number of daily fines is determined by the nature of the crime.

Examples of traffic violations that can result in daily fines:

- Drunk driving.
- Reckless driving.
- Driving without a license.

Prison

Serious traffic violations can lead to imprisonment.

Examples of serious traffic violations are:

- Rough negligence.
- Aggravated drunken driving.
- Escaping from the scene of an accident with personal injury.

Study Questions

Study questions are answered using the information above including the **Read More** words available. External links are only in-depth.

Who can decide on a law that applies in traffic?

The Riksdag (Riksdagen)

The Transport Agency

The Transport Administration

The Government (Regeringen)

The cars impact on the environment

Cars and traffic impact the environment

In the early days, people did not think much about how the car affected the environment. For example, until the late 1980s, a law was passed on catalysts for cleaner exhaust gases.

A great deal is about the exhaust gases emitted from cars with internal combustion engines when driven. However, it is not enough to look only at fuel and exhaust gases; one has to consider account the whole "life cycle" of the car. You have to think about what material is used to make the car, how fuel is produced, and what happens when it is scrapped. Many aspects together affect the environment. It can be complicated to say exactly what applies as the development progresses fast. Many different interest groups emphasize different "truths," which can make the information difficult to interpret.

The number of cars also increases the problems for society in the form of congestion, noise, and accidents.

It is, therefore, important that you have good knowledge of how your choices affect the environment. This can make a big difference.

The environment and your choices

Although technological developments are constantly advancing, the environment is adversely affected by road traffic. On the Swedish Transport Administration's website, one can read that of the total carbon dioxide emissions in Sweden, about 30 percent come from the road transport sector.

You as a driver can do a lot to reduce the car's negative environmental impact:

- You can choose to drive as environmentally friendly a car as possible.
- You can choose to drive in a climate-smart way, so-called "eco-driving".
- You can skip the car sometimes and maybe use the bike for shorter trips.

Your choices make a difference.

No internal combustion engines from 2035

European governments have taken the historic decision to stop the sale of polluting cars. Transport is the biggest source of emissions and cars are the biggest part of the problem.

From 2035, no new cars with internal combustion engines will be allowed to be sold within the EU.

Environmental zones



Click on the picture to enlarge4584

What is an environmental zone?

An environmental zone is an area in a municipality where only certain vehicles are allowed to drive.

Why are there environmental zones?

Many cities today have problems with bad air. This lousy air is primarily due to pollution from traffic.

Who decides where there will be environmental zones?

It is up to each municipality to decide whether it wants to introduce environmental zones.

What kind of environmental zones are there?

- Environmental zone class 1 – only covers heavy trucks and heavy buses. More about the requirements for Environmental Zone Class 1 can be found on the Swedish Transport Agency's website.

- Environmental zone class 2 - includes passenger cars, light buses, and light trucks. To be allowed to drive in environmental zone class 2 with a vehicle with a spark-ignition engine (e.g. petrol engine), the vehicle needs to meet the requirements for Euro class Euro 5 or Euro 6. To be allowed to drive a vehicle with a compression-ignition engine (diesel engine) in environmental zone class 2, the vehicles must meet Euro 6.
- Environmental zone class 3 – covers electric vehicles, fuel cell vehicles, and gas vehicles regardless of whether they are light or heavy. Electric vehicles, fuel cell vehicles and gas vehicles are allowed to drive here, both light and heavy vehicles (for gas vehicles, Euro VI emission requirements apply). In the case of heavy vehicles, plug-in hybrids are also allowed to drive if the vehicle meets the Euro VI emission requirements.

How do I know it's an environmental zone?

If you look at the picture, you will see the four types of road signs that occur. As it is a zone mark, it applies until you pass an end mark.

Are there any exceptions?

Yes, for example, vehicles that drive for the transport service, or vehicles whose driver or passenger has a parking permit for the disabled can be exempted from the rules.

Examples of the use of environmental zones

Municipalities can have several zones. For example, in Stockholm, there is a class 1 environmental zone that covers the entire inner city and prevents heavy traffic. Then you have environmental zone class 2 in particularly exposed areas within this zone.

Electric cars and environment

Loading image...



Click on the picture to enlarge4335

An electric car that runs only on electricity does not emit any exhaust gases and in that sense it is clearly much better for the environment than a car powered by regular gasoline or diesel. This applies both when it comes to carbon dioxide emissions and other toxic exhaust gases and particles. If we look at the entire life cycle of the car and the production of electricity, it becomes more complicated. In general, it can be said that an electric car that runs on electricity is usually better for the environment than the corresponding car powered by regular fossil fuels, even in terms of the entire life cycle. But how much better is dependent on a variety of factors.

Electricity

What type of electricity is used to charge the car? In Sweden we have good conditions to drive environmentally friendly with an electric car since most of the electricity in Sweden is renewable by eg hydropower. In countries where much of the electricity is produced by, for example, burning coal, it is usually still better for the environment with an electric car, but the difference becomes smaller compared to a fossil-fueled car. Many countries are working towards more environmentally friendly production of electricity so this aspect will change over time.

Battery

How big a battery does the electric car have, how is the battery manufactured and what substances does it contain? One of the biggest environmental factors in an electric car is

the battery itself. This contains substances that are obtained through mining, which in turn can greatly affect the environment. The larger the battery the electric car has, the more substances need to be extracted from the mines. This part is also under rapid development and different manufacturers have different amounts of influence on the environment.

The life cycle of the car

How is the car manufactured and how is the car taken care of during scrapping? If the factory that makes the car is powered by, for example, coal or solar power also affects how environmentally friendly the car is as a whole. If an electric car battery can be disposed of after scrapping the car and used for, for example, storing electricity for houses or other activities, the "life cycle" is considerably extended and the overall environmental impact is reduced. These are also aspects that can be taken into account. And here too, as with most other aspects of the electric car, the development is moving forward fast.

The whole picture determines

It can be difficult as a consumer to get a grip on the whole picture. If you are going to buy or rent an electric car, it is important for both yourself and society at large that you familiarize yourself with the subject and make conscious choices. Ask questions and demand answers. Not least, car manufacturers are becoming increasingly aware that clear information is needed to the buyer about how their product affects the environment and how you can, with your choices, be able to influence the environment in a good direction.

[Elbil förening, Elbil i Sverige](#)

Fuel

Loading image...



[Click on the picture to enlarge](#)⁸¹

Most of the greenhouse gases emitted from the transport sector come from road traffic. Emissions from passenger cars and heavy vehicles dominate.

Fossil fuels

Fossil fuels refer to fuels derived from natural gas, coal, and oil (regular petrol and diesel). Fossil fuels increase the amount of carbon dioxide in the atmosphere when used. Fossil fuels are not renewable.

Non-fossil fuels

Non-fossil fuels are energy coming from the sun, wind, water, and biomass (for example, biogas, ethanol, and HVO).

When biogas and ethanol are used, it does not increase the amount of carbon dioxide in the atmosphere if one takes into account that the plants that form the basis of the fuel have absorbed carbon dioxide at an earlier stage.

Biofuel

Biofuels are a collective name for fuels produced from biomass, such as food waste and plant parts. Sustainably produced biofuels are often considered carbon dioxide neutral as the carbon dioxide emitted during combustion is constantly, in the longer term, bound to new biomass in a closed cycle.

There are many types of biofuels, here are some examples:

Ethanol is made from corn, sugar cane, forest waste. Ethanol is often used as admixture in gasoline.

Methanol made from the parts of trees that are not taken advantage of when logging. There are also special trees grown as so-called "energy forests". Methanol is unusual but can be used as a mixture in gasoline.

Biogas is produced by biological material degraded by bacteria. Note, do not confuse natural gas with biogas. Natural gas is a fossil gas.

RME/Rapeseed oil produced from rapeseed. There are already diesel engines that can run on fuel made from rapeseed without the need to change them.

HVO stands for "hydrogenated vegetable oils" and is a fuel with fossil-free raw materials designed to mimic and replace fossil-fueled diesel.

The biggest advantage for our environment when using biofuels is that they do not cause any carbon dioxide contributions since the original material consume carbon dioxide while growing up.

The use of biofuels is increasing in our society. A common mix available at most gas stations are E85, which contains about 85% ethanol and about 15% gasoline (However, in winter, vehicles may be difficult to start with E85 and the mixture may then change to about 70 percent ethanol and up to 30 percent gasoline).

fuel made from

fuel made from

[Utsläpp av växthusgaser, Naturvårdsverket](#)

There is a difference between "gas" and "gas"

Loading image...



Click on the picture to enlarge3127

Vehicle gas is a collective name for the propellants used in gas vehicles. Vehicle gas consists of biogas and natural gas in varying mixtures.

Biogas is renewable

Biogas is a renewable fuel that produces much less carbon dioxide emissions than gasoline and diesel. Biogas can be extracted by digestion of, for example, food waste, manure or sewage sludge.

Natural gas is fossil

Natural gas is a fossil fuel that is extracted from oil wells. Natural gas is not renewable. Driving on natural gas produces greater emissions of greenhouse gases than biogas, but lower than petrol and diesel.

Catalyst internal combustion engine

In 1989, it became obligatory by law to use a catalyst for gasoline-powered cars in Sweden. The catalyst cleans the exhaust gases from harmful substances and greatly reduces environmentally hazardous emissions. Now, most cars with internal combustion engines are equipped with catalytic converters, including diesel cars.

Purifies the exhaust gases from dangerous substances but does not reduce carbon dioxide emissions

The catalyst cleans the exhaust gases from dangerous substances by converting harmful substances into essentially carbon dioxide and water. The catalyst does not reduce carbon dioxide.

The **easy** way to reduce carbon emissions is to reduce the consumption of fossil fuels. For example, use more ecodriving and sometimes maybe abstain completely from taking the car.

Working temperature

The catalyst only works when it has reached its operating temperature. During cold starts and short journeys, the purification of exhaust gases will not work, and emission of hazardous substances will increase significantly. When the catalyst has reached its operating temperature, it can be up to 600 degrees centigrade. If the engine speed exceeds 3000 rpm, the catalyst works poorly.

Six hundred degrees is very hot and as the catalyst is located under the car there is a risk of fire when you park the car in the high and dry grass.

[Katalysatorn, Avgassystem.se](http://Katalysatorn.Avgassystem.se)

Environmentally friendly driving

To think and drive in an environmentally friendly way is to plan and drive in a way that does not damage the environment more than necessary. For this definition to not be mere words you, as a soon to be driver, should consider the following :

- What type of transportation will you use?
- What vehicle are you going to buy?
- How much are you going to drive?
- Will you use environmentally friendly driving?

If you compare car, air, train, and bus, in Sweden, the train is the transport that has the least impact on the environment. You can advantageously cycle shorter distances.

As you probably understand from the questions, your choices affect the environment.

Study Questions

Study questions are answered using the information above including the **Read More** words available. External links are only in-depth.

Loading image...



Click on the picture to enlarge202

How do you travel best if you want to affect the environment as little as possible?

With your own car

By plane

By train

Loading image...



Click on the picture to enlarge222

What is correct?

Petrol contain the substance benzene which is a carcinogen

Petrol is a renewable fuel

Petrol contains a lot of sulfur compounds which cause acidification of forests and fields

Loading image...



Click on the picture to enlarge204

Can you reduce the amount of environmentally harmful exhaust emissions during cold starts of an internal combustion engine?

Yes, by leaving the car at idle for a few minutes before I drive away

Yes, by driving on a low gear and high number of revolutions so that the engine gets hot as soon as possible

Yes, by starting on a high gear when I have stopped

Yes, by driving with a low number of revolutions and at a constant speed

Loading image...



Click on the picture to enlarge1030

Why is it good to use an engine heater?

With a engine heater a catalyst is not necessary

With an engine heater I avoid cold starts which in turn reduce harmful emissions

With an engine heater I do not need to drive in an enviromentaly friendly way

What does the E10 label mean when you refuel?

95 octane gasoline with 10 % ethanol mixed

98 octane gasoline with 10 % ethanol mixed

If you are only going a short distance, what is best for the environment?

Taking the bus

Take the bike

Taking the local train

Take the car

Loading image...



Click on the picture to enlarge5536

What percentage of the total amount of carbon dioxide emissions in Sweden come from road transport sector?

About 10%

About 20%

About 30%

About 40%

About 50%

Which statement is correct regarding "biofuel"?

Biofuels are considered carbon neutral

Biofuels are the purest form of diesel oil

Biofuels are called E85, which means that it is 15 % ethanol in the gasoline

Which statement is correct regarding "natural gas"?

Natural gas is a fossil fuel

Natural gas does not contribute to carbon dioxide emissions

Natural gas is created from forest products

Loading image...



Click on the picture to enlarge9754

You drive a diesel car with emission class Euro classification 6. A street you want to drive into has the road sign in the picture posted, what do you do?

I choose another road as I am not allowed to drive a diesel car on this street

I choose a different road since Euro classification 2020 emissions class is needed to drive on this street

I continue as I meet the requirements to drive on this street

The car affects the environment

Traffic contributes a lot to our air pollution. This page deals with a number of different emissions and substances.

Exhaust fumes

Exhaust fumes contain large amounts of substances, some of which are more dangerous to humans and animals than others. Particles and soot from the exhaust fumes can cause

respiratory problems and create allergies in us humans. Certain substances in the exhaust gases are also carcinogenic.

Some examples of substances in exhaust fumes :

- Carbon oxide
- Carbon dioxide
- Nitrogen oxide
- Hydrocarbons
- Ground-level ozone
- Sulfur dioxide
- Soot and particles

Some examples of technologies and other measures

- A catalyst greatly reduces harmful emissions. The catalyst does **not** reduce the emissions of carbon dioxide which is the gas that contributes to the "greenhouse effect".
- The fuel today contains very little sulphur, which has reduced emissions of sulfur dioxide, which is a colorless and cough-inducing gas.
- Particle filters on diesel engines has reduced emissions of soot and particles often over 90%.
- Driving behavior affect how much exhaust fumes that are released. Above all your driving behavior affect emissions of carbon dioxide. Therefore, drive in an environmentally friendly manner.

Carbon monoxide

When designing a combustion engine it can not achieve a complete combustion of the fuel. Had this been possible the exhaust gases had been basically harmless. Carbon monoxide is formed by incomplete combustion of the fuel. This completely odorless gas is harmful to humans. Our bloods ability to take up oxygen is reduced, which makes us tired and distracted .

Do not confuse carbon monoxide to carbon **d**ioxide. Carbon monoxide is harmful to humans and carbon dioxide contributes to the greenhouse effect. The catalyst reduces emissions of carbon monoxide but not carbon **d**ioxide.

Carbon monoxide (SV: Koloxid) is also called carbon monoxide (SV: Kolmonoxid).

Driving in tunnels, garages and other enclosed spaces

Keep in mind that the risk that you breathe in carbon monoxide increases when driving in areas with poor ventilation, such as tunnels and underground parking.

Carbon dioxide

Carbon dioxide is a colorless and odorless gas at room temperature. It is formed when both fossil fuels and biofuels are burned.

In normal amounts, carbon dioxide is harmless to us humans and is part of the air we breathe. Carbon dioxide is the factor that contributes most to the greenhouse effect. When we emit carbon dioxide, the earth's average temperature increases. With higher average temperatures, water vapor, which is the most important greenhouse gas, also increases.

The above is one of several reasons you must learn to drive environmentally friendly, thereby contributing to lower carbon dioxide emissions.

The standard fuel, gasoline or diesel, sold in Sweden generates somewhere between 2 and 3 kg of carbon dioxide per liter of spent fuel. These figures apply to standard fuels. There are many different types of fuels with other numbers today, but these fuels are not in the majority. The fact that the emissions weigh more than the fuel itself is because the carbon dioxide that is created contains large amounts of oxygen that are "taken" from the atmosphere.

Greenhouse gas emissions

Emissions from transport in road traffic account for a third of Sweden's total emissions of greenhouse gases.

Emissions from domestic transport come from:

- passenger cars,
- light and heavy trucks,
- buses,
- mc and mopeds,
- train,
- domestic flights and shipping.

Carbon dioxide emissions at different speeds

Lower speed is important

On a vehicle with an internal combustion engine, fuel consumption is affected by the speed you keep. The faster you drive, the more fuel it consumes. For every 10 km/h higher speed over 70-75 km/h you drive, fuel consumption increases by 0.5-1.0 deciliters per 10 kilometers.

Below are two examples based on a petrol-driven car with a fuel consumption of about 0.8 liters/Swedish mile.

- If you drive 100 kilometers at 110 km/h instead of 90 km/h, fuel consumption and carbon dioxide emissions are increased by 10-20%.
- If you drive 100 kilometers at 90 km/h instead of 110 km/h, fuel consumption and carbon dioxide emissions are reduced by 10-20%.

Keep the speed limits

By not exceeding sign posted speed and driving sparingly, we can all contribute to reducing carbon dioxide emissions from road traffic.

Nitrogen oxide

Nitrogen oxides (nitrogen monoxide and nitrogen dioxide) are a group of gaseous pollutants consisting of nitrogen and oxygen. They are mainly formed when fuels are burned at high temperatures, such as in gasoline and diesel engines (diesel engines emit the most nitrogen oxides). During the combustion process, nitrogen monoxide is primarily formed, which is then oxidized to nitrogen dioxide in the air.

Nitrogen dioxide contributes to the formation of ozone and ground-level ozone, which is harmful to human health. Exposure to high levels can lead to breathing problems, lung damage, asthma, and other respiratory diseases.

Nitrogen dioxide can also contribute to acidification of soil and watercourses.

A car's catalytic can help reduce nitrogen monoxide emissions by converting it into less harmful substances.

Hydrocarbon

There are several types of hydrocarbons contained in the exhaust gases. Some of them are carcinogenic.

The catalyst reduces emissions of hydrocarbons.

Ground-level ozone

Ground-level ozone is formed in polluted air under the influence of sunlight. Pollutants such as nitrogen oxides, hydrocarbon and volatile organic substances from both natural and human sources contribute to the formation of ground-level ozone.

The ozone is long-lived in the atmosphere and travels long distances. A large amount of ozone and ozone-forming substances come to Sweden from other countries with the help of the wind.

Harmful effects

There is clear evidence of harmful effects of ozone on exposure for shorter periods, but also with longer exposure to lower levels. Irritation and effects on lung function, especially in asthmatics, are known at elevated levels. Ground-level ozone damages crops and forests.

Catalyst

An internal combustion engine's catalyst reduces the emissions of the substances required to form ground-level ozone.

Sulfur dioxide

Sulfur dioxide mainly exist in exhaust gases from diesel engines. Sulphur dioxide contributes to acidification in the form of " acid rain " when mixed with nitrogen oxides .

Sulphur emissions from road transport have declined sharply because today's fuel contains very little sulfur.

Study Questions

Study questions are answered using the information above including the **Read More** words available. External links are only in-depth.

Why are hydrocarbons in vehicle exhaust fumes particularly dangerous for humans?

They affect our cells and are carcinogenic

They affect our heart and vascular system

They affect our outer skin

Loading image...



Click on the picture to enlarge5527

What substance from the cars exhaust impairs your blood's ability to carry oxygen?

Nitric oxide

Carbon monoxide

Carbon dioxide

What is true of the following statements?

Our airways are affected in a negative way by particulates and soot from car exhaust

Sulfur dioxide is a colorless gas that poses no danger to humans

Carbon monoxide is a colorless, odorless and tasteless gas that is harmless to us humans

How much carbon dioxide is created when one liter of petrol (95 octane) is burned ?

Between 2 and 3 kg

Less than 0,5 kg

More than 4 kg



Click on the picture to enlarge5529

What type of engine emits the most nitrogen oxide ?

Petrol

Electric

Diesel

Hybrid



Click on the picture to enlarge5528

Which of the following statements is correct?

By using so-called "eco-driving" I can reduce the environmental negative effects of cars

By using so-called " eco-driving " I do not release any carbon dioxide

By using so-called " eco-driving " I reduce carbon dioxide emissions but increase my expenses for fuel

Why is ground-level ozone harmful?

Forests and crops are negatively affected

People's bone is negatively affected

People's eyesight is negatively affected



Click on the picture to enlarge5534

What is correct?

Exhaust gases from gasoline vehicles contain substances that harm nature but not people

Exhaust gases from gasoline vehicles contain hydrocarbons that can cause cancer

Exhaust gases from gasoline vehicles with catalytic converters contain no hazardous substances



Click on the picture to enlarge9766

You drive a gasoline-powered car . How much does the carbon emissions increase from your car when you are on a motorway driving at constant speed of 110 km/h instead of 90 km/h?

Emissions increase by around 10-20 % regardless of whether the car has a catalytic converter or not

Emissions increase by about 5 % if the car has a catalytic converter

Emissions are not affected at all if the car has a catalytic converter

Emissions increase by about 40 % if the car does not have a catalytic converter

Traffic noise

Traffic noise is a problem for many who live near the road.

Many municipalities try to limit noise by setting up noise protection between roads and residential areas. As a driver you have an obligation to drive so that you do not unnecessarily hinder or disturb. You should therefore always strive to drive as quietly and softly as possible.

Noise



Click on the picture to enlarge85

Noise is a major problem for many. It is not very nice to hear someone drive aggressively with bad muffler when one might be trying to put their baby to sleep. Some people get headaches and concentration problems of high traffic noise.

Lower noise levels

As a driver, you must contribute to that the noise level is reduced and this can be done by your driving style and by how you maintain your vehicle .

A few simple precautions :

- Do not rev your engine unnecessarily when standing still.
- Do not accelerate unnecessarily.
- Make sure the silencer works well.

Noise barriers

Sometimes there are special noise barriers between residential areas and roads. These noise barrier reduces but does not completely eliminate the noise. When there are noise barriers, drive with extra care since the plank informs you that it is an area that has problems with noise from traffic.

Silencer

The vehicle's silencer is part of the exhaust system. Its function is, as its name suggests, to silence the sound.

Unfortunately, it happens that this part rusts or breaks, which results in an unnecessarily high noise level.

Virtual engine sound

Electric cars traveling at low speeds emit very little noise. So little that it could be a traffic hazard for children, pedestrians, cyclists and animals. This is especially true in low speeds, for example in parking lots.

The virtual engine sound system generates an engine sound to create a sound that can be heard as an engine noise when the car is running on electricity alone.

Law within the EU

All electric and hybrid cars sold in Europe from 2021 must have an audio device that automatically starts at speeds lower than 30km/h.

Study Questions

Study questions are answered using the information above including the **Read More** words available. External links are only in-depth.

Loading image...



Click on the picture to enlarge116

What is the main reason for the walls, often built around residential areas and busy roads?

You want to prevent cars driving into the area by going off road

They want to reduce noise for those who live in the area

They want to prevent animals from getting out on the road

They want to prevent children from playing near the road, outside area

Why should you not use a car with a defective muffler?

Other road users may be disturbed

Exhaust emissions increase

The car can be damaged

The car's surroundings (and disadvantages)

For some, the car is a "necessary evil". For some, it is a prerequisite for coping with everyday life. Some have the car as their biggest hobby.

The car has therefore been given a natural place in our lives. But we must learn to deal with the car's negative effects on our environment in the form of noise and emissions, among other things.

The Greenhouse effect

Natural process

The so-called greenhouse effect is a natural process where the earth's atmosphere captures and retains heat from the sun's radiation. Normally, the Earth emits as much energy as it receives from the Sun, keeping the temperature of the Earth at a level habitable for life. The atmosphere acts as a blanket that captures part of the outgoing heat from the Earth and prevents it from escaping into space.

Heating

The problem arises when human activities, such as the burning of fossil fuels and deforestation, lead to increased emissions of greenhouse gases such as carbon dioxide, methane and nitrous oxide. These gases increase the density of the natural atmospheric cover and result in increased heat trapping in the atmosphere. This leads to a global warming effect, resulting in climate changes such as rising sea levels, changes in rainfall patterns, melting glaciers and extreme weather conditions.

Conflict

Basically, the greenhouse effect is a natural and necessary process to keep the Earth's temperature at a level that supports life. The problem arises when human activities increase the amount of greenhouse gases in the atmosphere, amplifying this natural process and causing an imbalance in the Earth's climate system.

Climate change

Road traffic accounts for the large greenhouse gas emissions in the transport sector, and a large proportion of transport will continue to be on the road. Road traffic's climate-impacting emissions can be reduced by three strategies: vehicles with lower emissions per kilometer, increased share of renewable fuels, and reduced road traffic work.

Sweden's climate target

Sweden's climate target states that emissions from domestic transport, excluding domestic flights, should be reduced by at least 70 percent by 2030 compared to 2010. By 2045, Sweden shall have no net greenhouse gas emissions to the atmosphere.

Acidification

Acidification in Sweden has decreased. Despite this, lakes, streams and woodland are still considered to be acidified due to human activities. What causes acidification is, among other things, emissions from transport, energy plants, industry and agriculture.

Sulfur dioxide, nitrogen oxides and ammonia are the substances most important for acidification.

[Luftföroreningar och dess effekter - Naturvårdsverket](#)

Eutrophication

Eutrophication affects land as well as lakes, streams and seas. The problems are mainly found in southern Sweden, but eutrophication occurs throughout the country.

Eutrophication causes the vegetation to change gradually, and species adapted to nutrient-poor environments are displaced.

Eutrophication is caused by excessive levels of nitrogen and phosphorus in soil or water. These nutrients can end up in the environment through emissions to air of, for example, nitrogen oxides from car traffic, shipping and power plants. Other causes of eutrophication are leaks from agriculture as well as emissions from wastewater treatment plants and industries.

[Luftföroreningar och dess effekter - Naturvårdsverket](#)

Road pavement

Particles and hydrocarbons are toxic to humans and animals. They represent serious health problems in our traffic environment.

It is mainly wear from road pavement and tires and road dust that accounts for elevated levels of hazardous substances. Increased use of studded tires in our largest cities is also contributing. Emissions of heavy metals and volatile organic substances are mainly caused by the brakes of the cars.

Our health is affected by the air we breathe.

Road Salt

Under normal conditions, snow and ice melt at 0 degrees Celsius. But with the help of a little salt you can melt snow all the way down to negative 18 degrees C. The salt can thus create snow- and ice-free roads even when it is minus degrees.

Negative effects of road salt

The Swedish Transport Administration and other road administrators are aware that salt has negative environmental effects. Salt can affect vegetation and water closest to the road. Salt can also increase the problem of rust. But to combat the slippage on high-traffic roads, there are currently no alternatives that have less impact on the environment.

Rubber from tires

It is not only exhaust emissions that create problems.

Vehicles also create many other toxic substances, such as rubber dust from tires. A large part of this ends up in our waterways and in the air we breathe.

Study Questions

Study questions are answered using the information above including the **Read More** words available. External links are only in-depth.

Loading image...



Click on the picture to enlarge217

What substance contributes most to the acidification of our land?

Carbon dioxide

Carbon monoxide

Nitrogen oxides

Loading image...



Click on the picture to enlarge5535

Which statement about car wash is correct?

At a car wash there are systems for wastewater which reduces emissions of dirty water

When I wash the car on the street the dirty water is always cleaned by the municipal sewage treatment plant

Ecodriving step by step

It is in the interest of society that you drive with as little fuel consumption as possible. This page summarizes something called Eco Driving (Sparsam körning)

In your exercise run, from now on, you should always "eco drive". Then it becomes a habit. Research shows that you arrive faster while saving fuel.

During the driving test, make sure that you use a driving style that results in low fuel consumption.

Think ahead

Think ahead all the time, where are you going? Avoid hard braking and stopping. Keep a distance from the car in front. Plan your itinerary so that you avoid heavy traffic.

Manual shifting

Use the engine brake by simply releasing the throttle, do not depress the clutch and, try to roll towards the intersection or circulation area. Strive to have the right speed without having to brake.

Automatic transmission

Release the throttle and roll towards the intersection or roundabout. Depending on the type of car you have, the car will roll or brake at different speeds. Strive to have the right speed without having to brake.

Shifting

Manual shift only

Shift up **early** and drive on as **high a gear** as possible. Only drive a few car lengths on first gear before you put in the second. Feel free to change from second to fourth or from third to fifth as this reduces the total consumption.

Even speed

Drive at an even speed and maintain speed limits. Keep the speed as even as you can. This is easier if you keep a good distance to the car in front of you.

Engine brake with internal combustion engine

Manual shift only

Engine brake as much as possible. Plan the drive and reduce the speed by reducing the throttle in time, you will get the best effect if you release the throttle completely. Shift down just before engine speed drops to about 1,200 rpm. If you do not have a tachometer on the car, you will learn to listen to the engine sound.

Idling

Do not idle with your car

Standing still with the engine running, so-called idle, can cost you money unnecessarily while releasing toxic substances and greenhouse gases. In most municipalities, idling is prohibited for more than a minute.

Roll

- On a downhill slope, you can let the car roll based on the car's conditions. Sometimes you do not need to give any gas at all.

- On an uphill slope, try to keep the throttle even. If possible, you can try to get a little extra speed before the hill to make it easier to get up.

Manual shift only

If you have gained a little extra speed in front of an uphill slope, you can often use a higher gear in the entire uphill slope and thus avoid downshifting.

Service the car

Service the car according to the owner's manual. Change oils and filters, make sure the engine is clean. Check the tire pressure at least every two refueling. Too low tire pressure increases fuel consumption.

Roof box and cargo

Air resistance and weight affect the car's fuel consumption. Remove roof box and roof railing when they are not in use. Empty luggage compartments on things you do not need to drive around with.

Study Questions

Study questions are answered using the information above including the **Read More** words available. External links are only in-depth.



Click on the picture to enlarge10250

How does the car's automatic cruise control affect fuel consumption?

Fuel consumption is not affected at all

Fuel consumption can increase on downhills

Fuel consumption can increase on uphill slopes

Loading image...



Click on the picture to enlarge²²

Which of the following will save most fuel?

I switch to a higher gear downhill

I release the gas in advance and avoiding braking as much as possible

I switch to a lower gear uphill

I use the cars automatic cruise control as much as I can

Loading image...



Click on the picture to enlarge146

Can the car's fuel consumption be affected by how you service your car?

No

Yes